6COSC023W – Final Project Report

Predicting the Tourist Arrivals After Covid-19 Outbreak in Sri Lanka Using Machine Learning

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Declaration

This report has been prepared based on my own work. Where other published and unpublished source materials have been used, these have been acknowledged in references.

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Abstract

Predicting tourist arrivals has been done previously by many researchers. However, tourist arrivals have never been predicted after the covid 19 outbreak. The aim of this research project is to analyze the effectiveness of predicting tourist arrivals after the covid 19 outbreak in Sri Lanka and design, develop and evaluate a web-based tourism demand forecasting system that can accurately predict the number of tourist arrivals after the covid 19 outbreak. Therefore, in this research study, we propose a web-based tourism demand forecasting system that can predict the number of tourist arrivals for the upcoming months accurately. It integrates with two Power BI dashboards in order to gain deeper business insight and improve decision-making process efficiency. This system will allow users to perform forecasting using different statistical models, considering the time series as the baseline model, and trying to improve the forecasting accuracy using machine learning techniques. This research project was chosen using the agile software development approach to be more flexible to make changes at any moment. The mix approach was chosen as the appropriate primary data collection methodology for this project because data was gathered through interviews as well as questionnaires. The past tourist arrivals data from January 1977 to December 2021 was collected in order to build the models. The predictions were calculated from the ARIMA(3,1,1)(0,1,2)[12] model and NNAR(14,1,8)[12]. The error measurements of ME, RMSE, MAE, MPE, MAPE, MASE, and ACF1 are used to evaluate the models. This research study found that the NNAR(14,1,8)[12] model performs better than the ARIMA(3,1,1)(0,1,2)[12] model because it has comparably lower error measurement values than the ARIMA(3,1,1)(0,1,2)[12] model. Therefore, the NNAR(14,1,8)[12] model is the best model for forecasting tourist arrivals after the covid 19 outbreak.

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List of Abbreviations

SLTDA	Sri Lanka Tourism Development Authority
SCM	Sama Circular Model
DTF	Dynamic Transfer Function
SARIMA	Seasonal Auto-Regressive Integrated Moving Average
ARFIMA	Autoregressive Fractional Integral Moving Average
CART	Classification and Regression Tree
BPNN	Backpropagation Neural Network
KNN	K-Nearest Neighbor
MLR	Multiple Linier Regression
ANN	Artificial Neural Networks
RNN	Recurrent Neural Networks
ACF	Auto Correlation Function
MARS	Multivariate Adaptive Regression Splines
LBQ	Ljung-Box Q
ADF	Augmented Dickey-Fuller
MDA	Multiplicative Decomposition Approach
MAPE	Mean Absolute Percentage Error
MSE	Mean Square Error
MAD	Mean Absolute Deviation
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
ME	Mean Error
LL	Log Linear

1. INTRODUCTION

1.1 Problem Statement

Predicting tourist arrivals after the covid 19 outbreak is difficult because of the lack of data availability. On the other hand, it is also difficult to get an acceptable level of accuracy for the predictions (Volchek et al., 2019).

Predicting tourist arrivals has been done previously by many researchers. However, tourist arrivals have never been predicted after the covid 19 outbreak. Accurate research-based information on future tourism demand is particularly in high demand after the covid-19 outbreak because it has caused unprecedented challenges for tourism at the global level (Wickramasinghe and Ratnasiri, 2021). At this moment, the Sri Lanka Tourism Development Authority does not have a web-based tourism demand forecasting system that can predict tourist arrivals for the upcoming months. Currently, they are using a combination of manual trend analysis and time series analysis methods to make the tourist arrival predictions. However, this manual trend analysis method takes a long time, and they have to allocate the necessary number of members to perform different tasks in order to get these predictions. Therefore, this research project tries to address this problem by looking at the gap in predicting tourist arrivals after the covid 19 outbreak. Here, we propose a web-based tourism demand forecasting system that will automate the current tourist arrival forecasting process that has been done manually by the Sri Lanka Tourism Development Authority. If the process was automated, efficiency would be increased in the tourist arrival forecasting process, and it would also have a more productive effect on the operational management of the economy. On the other hand, this would be a great opportunity for government authorities to digitalize their business operations because they would be able to transform their key business processes using innovative technologies, which would result in delivering value for the government as well as the hotel management industry.

Further factors that are contributing to the main problem will be illustrated in **Figure 1.1.1**

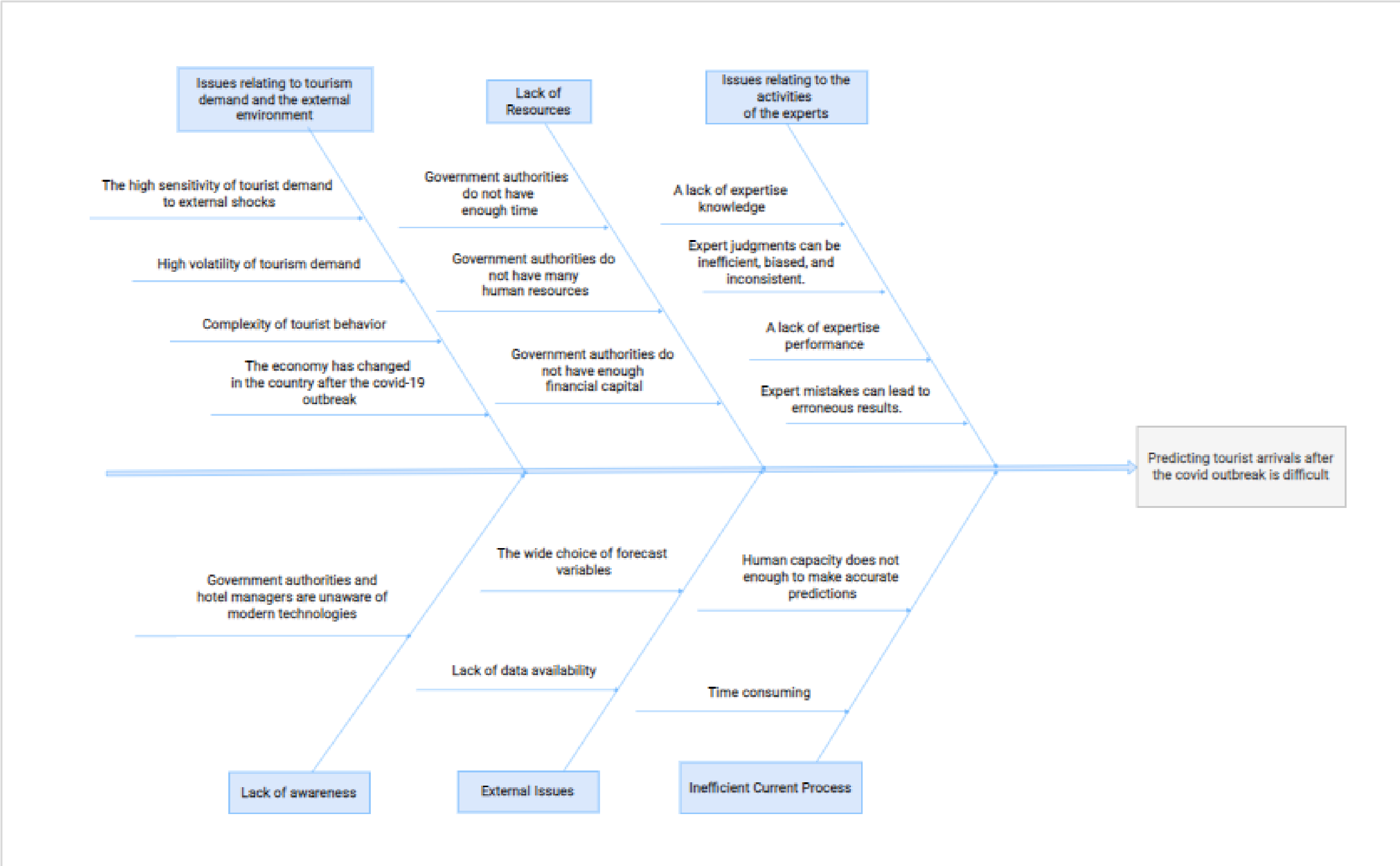


Figure 1.1.1: Cause and effect diagram

1.1.1 Issues relating to tourism demand and the external environment

Frechtling (2001) notes that the main challenges in tourism demand forecasting include the high sensitivity of demand to external shocks such as terrorism, earthquakes, floods, and diseases; the complexity of tourist behavior; and the high volatility of tourism demand (cited in Song, Gao and Lin, 2013, p. 1). In 2020, Sri Lanka's economy shrank by 3.6% (As Sri Lankan Economy Recovers, Focus on Competitiveness and Debt Sustainability Will Ensure a Resilient Rebound, 2021). It is an unprecedented economic downturn.

1.1.2 Lack of Resources

According to the details gathered from the Sri Lanka Tourism Development Authority (SLTDA), they do not have an automate process to predict the tourist arrivals. And also, they do have to allocate many human resources to predict the tourist arrivals and it takes a long time to make their predictions. On the other hand, they do not have enough time to predict tourist arrivals.

Furthermore, they do not have enough financial capital to purchase the tourism demand forecasting systems to make their predictions.

1.1.3 Issues relating to the activities of the experts

Song, Gao and Lin (2013) state that there is evidence that suggests the forecasters' judgements can be inefficient, biased, and inconsistent. In order to guarantee the accuracy of the predictions of the experts, they should reveal the exact probabilities, which are made up of a large number of estimations. On the other hand, the performance of experts is poor in both short- and long-term prediction domains (Epstein, 2019) because they have no sufficient knowledge about the prediction areas. Expert judgments are also less reliable since the world's most well-known specialists are rarely held accountable for their predictions. According to Tetlock, there is a negative correlation between expert forecasts and actual performance forecasting because they rarely accept systemic patterns in their decisions (Epstein, 2019).

1.1.4 Inefficient Current Process

SLTDA do have to follow three different scenarios, called "upper," "lower," and "average," in order to improve the accuracy of the predictions in their manual tourist arrivals forecasting process. As a result, this current forecasting process is time-consuming. Since they do not use any automate system at the moment, their predictions are almost based on human capacity, which is not enough.

1.1.5 External Issues

Frechtling (2001) notes that predicting tourist arrivals is difficult due to the lack of historical data availability and the wide choice of forecast variables, for instance, tourist arrivals, tourist expenditure, and hotel room nights (cited in Song, Gao and Lin, 2013, p. 1).

1.1.6 Lack of Awareness

Government authorities and hotel managers are unaware of modern technologies. The SLTDA, for example, currently bases its predictions only on time series analysis and manual pattern identification methods.

1.2 Delivered Aim and Objectives

1.2.1 Achieved Aim

The achieved aim of this research project was to analyze the effectiveness of predicting tourist arrivals after the covid 19 outbreak in Sri Lanka and design, develop, and evaluate a web-based tourism demand forecasting system that can predict the number of tourist arrivals accurately after the covid 19 outbreak.

1.2.2 Achieved Objectives

The following **Table 1.2.1** will review the achieved objectives concerning the objectives that have been set in the project.

# ID	Objective with Description	Status
	Literature Survey	
O1	Analyze the effectiveness of predicting tourist arrivals after the covid 19 outbreak.	Achieved
O2	Review the existing tourism demand forecasting systems and identify the gaps	Achieved
O3	Identify the models, techniques, and methodologies used in previous works and assess the benefits of using them in the proposed system	Achieved
	Requirement Analysis	
O4	Carry out an in-depth user requirement analysis and gather the system requirements in terms of what the user would like to see in the proposed system.	Achieved
	Design	
O5	Design the interfaces of the web-based tourism demand forecasting system based on the system requirements that are identified in the requirement analysis phase	Achieved
O6	Design two Power BI dashboards for past and predicted tourist arrivals and integrate them with the web-based tourism demand forecasting system	Achieved
	Development	

O7	Develop the web-based tourism demand forecasting system according to the designed prototype.	Achieved
O8	Implement the different statistical forecasting models.	Achieved
O9	Create the designed Power BI dashboards and integrate them with the web-based tourism demand forecasting system.	Achieved
	Testing	
O10	Performing functional testing to test the developed prototype	Achieved
O11	Performing user testing using the expertise of the Sri Lanka Tourism Development Authority and hotel management industry for evaluation.	Achieved

Table 1.2.1: Achieved Objectives

1.2.3 Timely completion of the project

In order to manage the project on time, the project schedule was created using a Gantt chart. With the submission of the summative assessments of Project Proposal (PP), Literature Review, Software Requirement Specification (SRS), and Project Specifications Design and Prototype (PSDP), I was able to gain a better understanding of the project flow. These deliverables of the projects which is presented in **Table 1.2.2** with the dates were also able to complete the project successfully on time. The following **Figure 1.2.1** will present the created Gantt chart for the project.

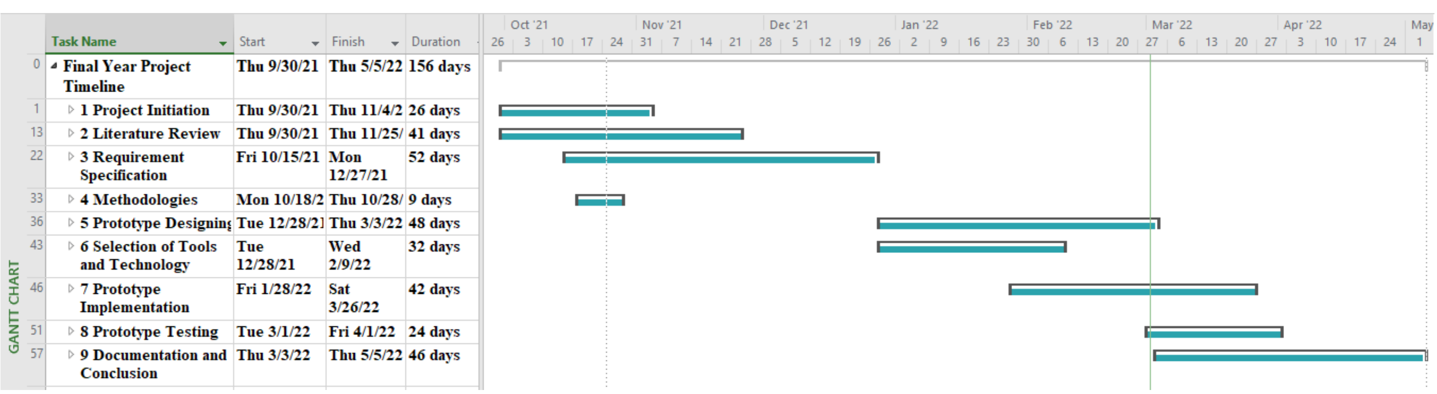


Figure 1.2.1: Gantt Chart

Please refer to **Appendix A – Detailed Gantt Chart**

Please refer to **Appendix B - Timeline**

Deliverable	Date
Project Initiation Document	4th of November 2021
Literature Review Document	25th of November 2021
Software Requirement Specification (SRS) Document	27th of December 2021
Project Specifications Design and Prototype (PSDP)	3rd of March 2022
Prototype Demo and Progress update	21st of March 2022
Draft Final Year Project Report	25th of April 2022
Final Year Project Report	5th of May 2022

Table 1.2.2: Project deliverables and dates

2. BACKGROUND

2.1 Literature survey

2.1.1 Literature Review of the domain

Tourism demand is the foundation on which all tourism-related business decisions ultimately rest, and the success of many businesses (airline companies, tour operators, hotels, cruise ship lines, and recreational facility providers) depends largely or totally on the state of tourism demand (Dwyer, 2007). The reason for that is the key role of demand as a determinant of business profitability and estimates of expected future demand constitute a very important element in all planning activities. Government authorities are the most beneficial parties in tourism demand forecasting since it is essential for them to successfully manage budget preparation, strategy development, and policy creation by predicting tourist arrivals using past data (Song and Li, 2008). On the other hand, tourism investments in destination infrastructures such as airports, highways, and rail links require long-term financial commitments, and the sunk costs can be very high if the investment projects fail to fulfill their design capacities (Dwyer 2007). Therefore, it is clear that tourism demand forecasting is essential to have a positive impact on the economy of the country.

2.1.2 Literature Review of the Sri Lankan Previous Research works

There are many studies that have been conducted previously in the field of predicting tourist arrivals. However, there has been relatively little research done in Sri Lanka to predict tourist arrivals before the covid 19 outbreak. For instance, the studies (Gnanapragasam, Cooray and Dissanayake, 2016 a) and (Gnanapragasam and Cooray, 2016 b) have proposed models for predicting international tourist arrivals after the civil war of Sri Lanka, respectively, using the State Space Modeling method and the Dynamic Transfer Function Modeling method. Two research studies (Peiris, 2016) and (Thushara, Su and Bandara, 2019) propose SARIMA model to predict international tourist arrivals to Sri Lanka. In the study (Peiris, 2016), the HEGY test was used to identify seasonality in the data set. This study states that the SARIMA (1, 0, 16) (36, 0, 24) 12 model is the suitable for forecasting tourist arrivals in Sri Lanka. The paper (Ishara and Wijekoon, 2017) uses time series modeling with the ARIMA method and the Multiplicative Decomposition Approach (MDA) to predict tourist arrivals in Sri Lanka. This study is conducted between January 2000 and February 2016 using two types of data (before the war and after the war). Ishara and Wijekoon state that post-war data produced more accurate results for the ARIMA model. The paper (Diunugala and Mombeuil, 2020) evaluates three different methods for predicting foreign tourist arrivals to Sri Lanka from the top 10 countries. According to the findings, Winter's Exponential Smoothing and ARIMA are the best approaches for forecasting tourist arrivals in Sri Lanka.

Wickramasinghe and Ratnasiri (2021) try to improve the accuracy of tourism predictions for Sri Lanka after the covid-19 outbreak using ARIMA, ARIMAX, SARIMA and SARMAX models, considering the number of tourist arrivals and guest nights. They state that SARMAX models are the most appropriate models prior to the other models. The study (Imam and Ananda, 2021) uses machine learning approaches such as CART, Boruta, hyperparameter tuning, grid search, innovative robustness check procedures, and Random Forest to find the tourism growth trends in Sri Lanka from 1972 to 2018. Imam and Ananda found that total tourist arrivals, tourism receipts, and seasonality affect the increase in both direct and indirect tourism employment. This research states that according to Random Forest models, a rise in tourist arrivals in the fourth quarter can more than compensate for any negative impact on direct and indirect employment growth caused by a fall in tourist arrivals and tourism earnings. The study (Nagendrakumar et al., 2021) aims to predict the tourist arrivals in Sri Lanka from August 2021 to August 2025 using past monthly tourist arrivals data from January 2000 to July 2021. Nagendrakumar et al. used Box-Jenkins ARIMA to predict tourist arrivals, and the Standard ARIMA model was used to achieve the purpose of the research. This study states that the results revealed that civil war has negatively impacted tourist arrivals.

2.1.3 Literature Review of the International Previous Research works

Similarly, many international research studies have been conducted on predicting tourism demand. For instances, the exponential smoothing method has been utilized in the research (Lim and McAleer, 2001) to predict quarterly foreign tourist arrivals in Australia. This study states that the Holt–Winters Additive and Multiplicative Seasonal models outperformed the Single, Double, and Holt–Winters Non-Seasonal Exponential Smoothing models in predicting. The paper (Cho, 2003) presents a study on predicting the number of tourist arrivals in Hong Kong from different countries using Exponential Smoothing, univariate ARIMA, and Elman's Model of Artificial Neural Networks. Cho states that neural networks appear to be the best method for forecasting tourist arrivals, especially for series with no evident trend. This research suggests that in the future, more complex forecasting techniques based on cutting-edge technology, such as Genetic Algorithms, Fuzzy Neural Systems, Probabilistic Rules, or other complex data mining techniques, should be used to predict tourist arrivals. The paper (Eleni and Matthew, 2010) proposes an advanced computational approach in order to predict non-scheduled or chartered international tourist demand using ARFIMA models. These ARFIMA models consider both non-stationarity and long-term memory effects in the auto-regressive process and temporal neural networks. This study reveals that the advanced form of temporal neural networks utilized in this research appeared to outperform the ARFIMA model in terms of prediction accuracy.

The paper (Song et al., 2010) introduces a new model, the TVP-STSM, with structural time series and a time-varying parameter regression approach in order to predict the tourist arrivals to Hong Kong. The study (Hadavandi et al., 2011) develops a Mamdani-type fuzzy rule-based system to predict tourist arrivals in Taiwan with high accuracy using a hybrid artificial intelligence (AI) model. This hybrid model used genetic algorithms for learning the rule base and customizing the database of the fuzzy system. Hadavandi et al. states that GFS forecasting accuracy is substantially better than other approaches, and GFS can be used as a viable forecasting tool for tourism demand forecasting problems. The paper (Lin, Chen and Lee, 2011) presents a study on predicting tourist arrivals to Taiwan using different models such as ARIMA, ANN, and multivariate adaptive regression splines. This study states that ARIMA is the best model for this scenario. Gounopoulos, Petmezas and Santamaria (2012) presents a study on short-term forecasts of tourist arrivals that have been generated in Greece. This study uses impulse response analysis to see how macroeconomic shocks from the source country would affect future tourism demand. Gounopoulos, Petmezas and Santamaria state that the ARIMA (1, 1, 1) model outperforms exponential smoothing models in forecasting the direction of one year of sample forecasts. The paper (Pai, Hung and Lin, 2014) presents a novel prediction system to predict tourism demand. To implement this system, Pai, Hung and Lin used fuzzy c-means with logarithm least-squares support vector regression technologies and stated that this system was able to achieve high accuracy performance.

The paper (Noersasongko et al., 2016) tries to predict tourist arrivals using BPNN, KNN, and MLR and then compares the performance of each. Noersasongko et al. state that BPNN performs well over the other algorithms. Yu et al. (2017) suggest a SA-D model that is implemented with the SARIMA and dendritic neural network models in order to predict tourism demand. This study states that this SA-D model achieved better predictive performance. In the study (Hassani et al., 2017), it evaluates the use of numerous parametric and nonparametric forecasting models in order to predict the arrivals of tourists in European countries. According to this study, no single model can produce the best short-, medium-, or long-term forecasts for any of the countries examined. Furthermore, forecasts from the Neural Network (NN) Model, Error, Trend, Seasonal (ETS) Model, Autoregressive Fractional Integral Moving Average (ARFIMA), Moving Average (MA) Model, and Weighted Moving Average (WMA) Model produce the least accurate predictions for European tourist arrivals. However, ARFIMA forecasts outperform the powerful NN, while MA and WMA forecasts exceed both ARFIMA and NN forecasts in certain circumstances.

Aydin and Yurdakul (2020) introduce a new framework using data envelopment analysis and machine learning algorithms that evaluates the performance of different types of countries during the covid-19 outbreak. The clustering analysis was performed using K-means and hierarchic clustering methods. The parameters are analyzed using the decision tree and random forest techniques. According to Aydin and Yurdakul, the optimum number of clusters for these different countries is three. The paper (Abdualgalil and Abraham, 2020) presents a study on predicting the number of tourist arrivals in India using machine learning algorithms. Here, Abdualgalil and Abraham use machine learning algorithms such as SVM, Naive Bayesian, Logistics Regression, Random Forest, Decision Tree, KNN, and SVR to make the predictions.

The study (Prilistya, Permanasari and Fauziati, 2021) examines the impact of the covid-19 pandemic and search query data on the prediction of international tourists to Indonesia using ARIMAX and SARIMAX models. However, the performance of the two models is then compared with the ARIMA and SARIMA methods. Prilistya, Permanasari and Fauziati state that the best accuracy is obtained by the SARIMAX method with Google Trends. Furthermore, this study creates a framework to build a composite search index for Google Trends to improve forecasting accuracy. Makoni, Chikobvu, and Sigauke (2021) use hierarchical forecasting methods to predict foreign tourist arrivals in Zimbabwe. These hierarchical forecasting methods include top-down, bottom-up, and optimal combination approaches. This study states that the bottom-up approach has shown the best results. The paper (Zhang et al., 2021) uses a combination of econometric and judgmental methods in order to predict the possible paths to tourism recovery in Hong Kong. This study uses the autoregressive distributed lag-error correction model to make baseline predictions, and Delphi techniques are used to perform different scenarios of analysis.

The paper (Yang, Guo and Sun, 2021) discusses predicting the tourism demand and the search behavior of tourists through segmented Baidu search volume data using econometric and machine learning models to improve its performance. This study states that the search behaviors of tourists have changed significantly with the prevalence and popularization of 4G technology. Furthermore, it proposes that search volume can evaluate the performance of the predictions, especially search volume on mobile terminals, from 2014M1–2019M12. The article (Höpken et al., 2021) proposes a method for improving autoregressive prediction models by integrating travelers' web search traffic as an external input characteristic for predicting tourist arrivals. This study presents a unique method for identifying relevant search phrases and aggregating them into a compound web-search index, which can then be utilized as a supplement to an autoregressive prediction methodology. Höpken et al. states that Google Trends data significantly improves tourist arrival prediction performance when compared to autoregressive approaches based solely on past arrivals, and that the machine learning technique ANN can outperform ARIMA models.

2.2 Review of projects / applications

2.2.1 Review of existing Tourism Demand Forecasting Systems

A tourism demand forecasting system (TDFS) is a forecasting support system that can generate quantitative tourism demand forecasts as well as allow users to create their own "what-if" scenario forecasts (Song, Witt and Zhang, 2008).

Nikolopoulos et al. (2003) note that research on web-based tourism demand forecasting systems is limited (cited in Song, Witt and Zhang, 2008, p. 452). The research studies (Song, Witt and Zhang, 2008, Song, Gao and Lin, 2013) develop web-based tourism forecasting systems to predict the tourism demand in Hong Kong using the VAR model and the ADLM model, respectively. Patelis et al. (2005) presented a tourism demand forecasting system in an e-government context where government information was available on the Internet (Song et al., 2008). Several national/regional tourism organizations, as well as international organizations such as the World Tourism Organization, the World Travel and Tourism Council, and the Pacific Asia Travel Association, have developed tourism forecasting systems that provide useful information to policymakers at various levels. Other countries, such as New Zealand's Ministry of Tourism and Australia's Tourism Forecasting Council, produce tourism forecasts by combining quantitative forecasts with judgmental adjustment (cited in Song, Witt and Zhang, 2008, p. 448).

Based on the existing applications, the research gap is identified as follows,

- There is no existing web-based tourism demand forecasting system solution in the Sri Lankan context.
- The tourist arrivals predictions of the existing web-based tourism demand forecasting system have not been updated after the covid 19 outbreak. As a result, there have been accuracy issues with the predictions in these existing systems.
- There are some functionality issues in the existing web-based tourism demand forecasting system. For instance, we are unable to log into the Hong Kong Tourism Demand Forecasting System after register to the system.

- These existing systems are unable to forecast tourist arrivals in Sri Lanka.
- Existing web-based tourism demand forecasting systems use a single model to perform statistical forecasting.
- Although a dashboard is available in a few existing applications, it is unable to give both past and future details in a useful manner, as well as it cannot be customized further.

Therefore, in this research study, we propose a web-based tourism demand forecasting system for the Sri Lankan context that can predict the number of tourist arrivals for the upcoming months accurately. It integrates with two Power BI dashboards in order to gain deeper business insight and improve decision-making process efficiency. This system will allow users to perform forecasting using different statistical models, considering the time series as the baseline model, and try to improve the forecasting accuracy using machine learning techniques. Furthermore, it will allow users to view and download customized reports of past and predicted tourist arrivals.

The feature comparison of the existing web-based tourism demand forecasting systems has been performed in the following **Table 2.2.1**.

Feature	Hong Kong Tourism Demand Forecasting System	Australia International Tourism Forecasts	Tourism Prediction System	Tourism Statistics | World Tourism Organization	International Tourism Forecasts | New Zealand Ministry of Business, Innovation & Employment	Sri Lanka Tourism Demand Forecasting System
Login and Register	√	×	×	√	×	√
Ability to select a time range	√	√	√	×	×	√
Ability to make predictions based on the number of months they want to forecast	×	×	×	×	×	√
Ability to get the predictions from different statistical forecasting models	×	×	×	×	×	√
Ability to get the predictions of the tourist arrivals based on the inputs selected	√	√	√	×	×	√

Ability to view the predicted tourist arrivals reports	×	√	×	×	√	√
Ability to customize the predicted tourist arrivals reports.	×	×	×	×	×	√
Ability to download the predicted tourist arrivals reports	×	√	×	×	√	√
Ability to download the predicted tourist arrivals reports in both excel and pdf format.	×	×	×	×	×	√
Ability to adjust the predictions	√	×	×	×	×	√
Ability to view predicted tourist arrivals on a dashboard	×	√	×	√	×	√
Ability to view past tourist arrivals on a dashboard	×	×	×	×	×	√

Ability to customize the dashboards	×	×	×	×	×	√
Ability to provide comments and suggestions	√	×	×	×	×	√
Ability to receive notifications	×	√	×	×	√	√
Ability to view customized past tourist arrivals reports.	×	×	×	×	×	√
Ability to download the customized past tourist arrivals reports in both excel and pdf format	×	×	×	×	×	√

Table 2.2.1: Feature comparison

2.2.2 Review of algorithms and techniques used in previous research

Tourism demand forecasting can be done using either quantitative or qualitative approaches (Dwyer, 2007). These approaches will be explained in the following section.

Quantitative forecasting approaches

The quantitative forecasting approach is usually used in short-term predictions by using past data to predict future data with numerical data and continuous patterns (Sona, 2019). Time series analysis models, econometrical models, and machine learning models are the most widely used quantitative forecasting approaches in the previous studies. Time series models can be classified as traditional time series models and machine learning models. Traditional time series models can be further categorized into univariate and multivariate depending on the number of values that they predict. Univariate models are used to forecast the outcome of a single time series. These models are the ARIMA model and the SARIMA (seasonal ARIMA) model, which have been used in many studies of tourism demand forecasting (Cho, 2003; Diunugala and Mombeuil, 2020; Gounopoulos, Petmezas and Santamaria, 2012; Höpken et al., 2021; Ishara and Wijekoon, 2017; Konarasinghe, 2021; Lin, Chen and Lee, 2011; Peiris H., 2016; Prilistya, Permanasari and Fauziati, 2021; Thushara, Su and Bandara, 2019; Wickramasinghe and Ratnasiri, 2021; Yu et al., 2017 and so on). However, the time series analysis has several weaknesses, including difficulty generalizing results from a single study, difficulty getting proper measurements, and difficulty accurately finding the best model to represent the data (Velicer and Plummer, 1998). On the machine learning front, time series analysis can be performed with almost any type of regression model. It could be a neural network regressor, which consists of neural networks with a single output neuron that determines the label of the regression. The predicted variable in econometric forecasting is connected with a set of deciding factors that are combined with the predicted quantitative association between the predicted variable and its determinants to produce future values of the predicted variable (Dwyer, 2007). Few attempts have been made to predict tourism demand using econometric forecasting models (Dwyer, 2007). For instance, the Vector AutoRegressive (VAR) model used in the study (Song, Witt and Zhang, 2008). AI models can capture nonlinear correlations and patterns across time series and exogenous variables in tourism demand forecasting. As a result, AI models allow them to improve forecasting performance. Before modeling, non-parametric AI models did not require explicit specification or data probability distribution. However, before implementing time series and econometric models, the chosen model's specification and the probability distribution for data must be assumed. As a result, non-parametric AI models outperform time series and econometric models, particularly when dealing with non-normal and nonlinear data (Jiao and Chen, 2019). The papers (Cho, 2003; Hassani et al., 2017; Höpken et al., 2021; Lin, Chen and Lee, 2011; Noersasongko et al., 2016; Yu et al., 2017) use neural network models to perform tourism demand forecasting. The paper (Abdualgalil and Abraham, 2020) has described the different types of machine learning techniques in order to predict number of tourists arrived in India using algorithms such as Support vector machine (SVM), Naive Bayesian, Logistics Regression, Random Forest, Decision Tree, K-Nearest Neighbors (KNN) and Support Vector Regression (SVR).

Qualitative forecasting approaches

The qualitative method, which is also known as the judgmental method, offers subjective results as it is comprised of personal judgments by experts or forecasters. Qualitative forecasters are often biased because they are based on an expert's knowledge and experience and rely on data. Qualitative (survey) methods provide results that are generally more descriptive and detailed than quantitative methods. Among the qualitative methods such as sales force opinion, Delphi techniques, and market research, the Delphi technique is one of the most popular quantitative methods used in tourism demand forecasting. Here, each expert is responsible for generating a forecast for a particular segment assigned to them. The team further discusses the forecast a consequent forecast is made by the team again and the process is repeated until all the experts reach each a general agreement. The Delphi technique has become a widely used decision-making tool in a wide range of fields (Rowe and Wright, 1999). The study (Zhang et al., 2021) used the Delphi technique and scenario analysis to generate current forecasts of tourism demand recovery in response to unexpected effects of disasters. The study (Song, Witt and Zhang, 2008) has used the Delphi method for the second stage of judgmental adjustment in their proposed web-based tourism demand forecasting system.

Forecasting Evaluation

In previous work, the forecasting performance assessment has been done using different evaluation measurements. Among them, the most commonly used performance measures are Mean Absolute Percentage Error (MAPE), which has been used many times (Cho, 2003; Gnanapragasam, Cooray and Dissanayake, 2016 a; Diunugala and Mombeuil, 2020; Gnanapragasam and Coorary 2016 b;

Ishara and Wijekoon, 2017; Konarasinghe, 2021; Makoni, Chikobvu and Sigauke, 2021; Prilistya, Permanasari and Fauziati, 2021; Wickramasinghe and Ratnasiri, 2021; Yang, Guo and Sun 2021 and so on), Mean Absolute Error (MAE), which has been used in the studies (Aydin and Yurdakul, 2020; Peiris H., 2016; Prilistya, Permanasari and Fauziati, 2021; Wickramasinghe and Ratnasiri, 2021; Yang, Guo and Sun, 2021 and many more) and Root Mean Squared Error (RMSE) which has been used in the papers (Aydin and Yurdakul, 2020; Cho, 2003; Höpken et al., 2021; Peiris H., 2016; Prilistya, Permanasari and Fauziati, 2021; Wickramasinghe and Ratnasiri, 2021; Yang, Guo and Sun 2021 and so on). Other evaluation measures include Mean Squared Error (MSE), Mean Absolute Deviation (MAD), which have been used in the studies (Konarasinghe, 2021), Normalized Mean Square Error (NMSE), Absolute Percentage of Error (APE), Correlation Coefficient (Wickramasinghe and Ratnasiri, 2021), and Program Running Time (PRT) which has been utilized in the paper (Yu et al., 2017).

The following **Table 2.2.2** will describe the advantages and disadvantages of the methods and algorithms that were used in the previous work.

Methods/ Algorithms	**Advantages**	**Disadvantages**
ARIMA	- Simple model - Ability to handle the autocorrelation embedded in the data	- Missing values can have a significant impact on model performance - Unable to recognize complex data patterns - Usually only useful for short-term forecasts, not for long-term forecasts
Exponential smoothing	- Simple to understand and put into practice - Producing accurate forecasts for short-term forecasts - Gives more significance to recent observations	- Generating forecasts that are lag behind the actual trend - Inability to handle trends well
RNN	- Missing values have no substantial impact on the performance of the model	- When trained on long time series, RNNs typically suffer from the vanishing gradient or exploding gradient problem

	- Capable of detecting complicated patterns in time series input - Giving best results in forecasting with few-steps - Ability to model sequence of data	- Have a poor memory and unable to incorporate several elements from the past into future prediction - The training of a Recurrent Neural Network is hard to parallelize and is also computationally expensive
VAR	- Flexible and less demanding in information and time - Allowing to integrate new data easily	- Lack of description and economic explanation of the linkages between variables

Table 2.2.2: Methods and algorithms of the previous research and their advantages and disadvantages

2.3 Review of tools and techniques

2.3.1 Review of tools

The following **Table 2.3.1** will discuss the advantages and disadvantages of the tools that are relevant to the project.

Tool	Advantages	Disadvantages
Programming Languages and Environments		
R	- Open-Source language - Statistical language - Allowing to perform data wrangling - Array of Packages - Quality Plotting and Graphing - Ability to run the code on all operating systems - Allowing to do various machine learning operations - Continuously Growing	- Not supported for dynamic or 3D graphics - Requires more memory as compared to Python, so that does not suit when dealing with big data - Basic security is lacking. As a result, R has a lot of limitations because it can't be embedded in a web application. - Complicated Language - Slower than other programming languages (MATLAB and Python)

RStudio	- Easy to write scripts - Convenient to view and interact with the objects stored in the environment - Easy to set the working directory and access files on the computer - Making graphics much more accessible for a casual user	- Processing is slow when working with large datasets
Shiny framework	-Excellent for quick prototyping - Easy to use - Useful for creating dashboards and websites that display a variety of metrics in an interactive manner - Ideal for quick data entry - Excellent for more advanced statistical modeling and visualizing the results	- Apps with multiple pages are not recommended - No async support (but in progress) - It is challenging to use for gathering and saving data to the database - Maintaining an application costs money - When the software expands in size, the code structure will change
Python	- Easy to use code lines, excellent handling and debugging - Providing large standard libraries - Ability to integrate with the Enterprise Application - Having extensive support libraries and clean object-oriented designs	- Weak language for mobile computing - Memory consumption is very high - Python executes with the help of an interpreter instead of the compiler, which causes it to slow down - Requires more testing time, and the errors show up when the applications are finally run
Database		
MySQL	- Open-source relational database - Supporting for programming languages – Java, Python, Node.js, Server-side PHP	- ROLE, COMMIT, and stored procedures are not supported in MySQL versions below 5.0. - MySQL does not support very large database sizes

	- Allow organizations to distribute, modify or deploy cloud-native applications - To connect to a MySQL server, a variety of secure and seamless connection options are available - Ability to use in popular web applications (WordPress, Drupal, Joomla, Facebook and Twitter)	- MySQL is inefficient in handling transactions and is prone to data corruption - In comparison to other databases, MySQL lacks a good development and debugging tool - SQL check constraints are not supported by MySQL
MongoDB	- Non-relational database management system. - In comparison to SQL queries, MongoDB's document query language is quite easy - Supporting for programming languages (Java, Python, Node.js, Server-side PHP) - Allow organizations to distribute, modify or deploy cloud-native applications Do not need to design the schema of the database	- Using high memory of data storage - Including document size limits - Transactions are not supported by MongoDB
Data Visualization		
Power BI	- Ease of use - When the amount of data is limited, it is faster and performs better - Ability to integrate with other Microsoft products easily - Easy to learn and understand the Power BI interface - Supporting for various data resources	- When working with large amounts of data, Power BI tends to slow. - Limited access to other databases and servers - Microsoft revolution analytics is only available to enterprise users

	- Supporting for data analysis expression and M language for data manipulation and data modeling - Creating reports and dashboards with a wide range of detailed and attractive visualizations - Ability to ask questions about data and obtain meaningful insights	
Tableau	- Ability to handle large volumes of data quickly - Faster and has a lot of capabilities for visualizing data - Enable to create and customize the dashboards according to user requirements using its intelligent interface - Ability to access numerous data sources and servers - Integrating much better with the R language - Tableau Software Development Kit can be implemented using C, C++, Java, and Python - Ability to customize dashboards for a device - Supporting Python machine learning features	- High cost - Inflexible Pricing - Poor After-Sales support - Security Issues - Poor BI capabilities - Embedment Issues

Table 2.3.1: Relevant tools with their advantages and disadvantages

3. LEGAL, SOCIAL AND ETHICAL ISSUES

3.1 Legal Issues

User data will not be exposed to external parties under any condition in this project, and only the registered users of the system will be able to access the system data. All the past tourist arrivals data that was used to train the model was gathered from the online public access websites of the Sri Lanka Tourism Development Authority (https://sltda.gov.lk/en) and Trading Economics (https://tradingeconomics.com/sri-lanka/tourist-arrivals). Furthermore, the system was built with the GPL-3 license in mind, and all tools, languages, and packages are licensed under open-source licenses. Therefore, it is clear that there are no issues involved with this project in terms of legal.

3.2 Social Issues

Even though this proposed web-based tourism demand forecasting system tries to automate the current manual tourist arrivals forecasting process that has been followed by the Sri Lanka Tourism Development Authority, the authority's employees who are not adapted to the technology or lack knowledge and no prior experience with the technology might be refused to use this system. As well, hotel managers who are not willing to adapt to the technology might also not use this system. In that sense, there is a possibility that some social issues may arise.

3.3 Ethical Issues

All the participants were well informed about the purpose of collecting data before gathering the system requirements from them. As well, the evaluators have been notified, and after that, their feedback has been included in the research project. Based on the target audience, both questionnaires were created in English and Sinhala to avoid the problems that can occur due to language barriers. Therefore, there is no ethical issue involved with the project.

3.4 Professional Issues

With the automation of the manual forecasting process of tourist arrivals, the tourism forecasting experts who make the predictions manually might think this system will be a reason for their skills being pointless. And also, the employees who do not have the knowledge and experience to use this system could be replaced by their profession.

4. DESIGN

4.1 System Design of the Prototype

The system design of the project has been illustrated in **Figure 4.1.1.**

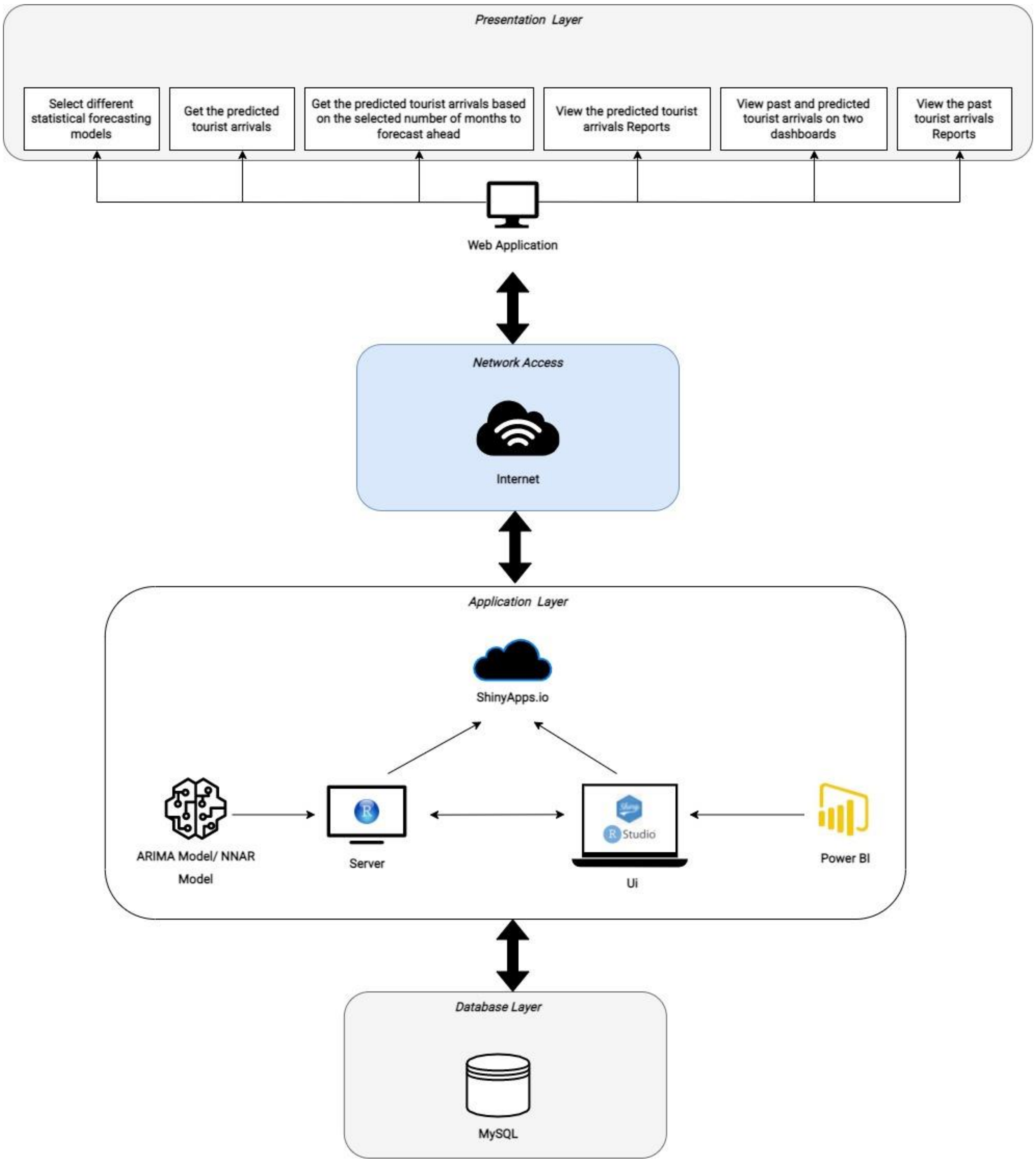


Figure 4.1.1: System design of the prototype

4.2 Final List of Requirements

4.2.1 Functional Requirements

Finalized Functional requirements have been presented in the following **Table 4.2.1**.

Req ID	Functionality	Priority	Use Case
FR1	Unregistered users shall be able to create an account to register with the system.	M	Register
FR2	The system shall be able to validate the required user inputs when creating a new account.	M	Validate user inputs
FR3	The system shall be able to display error messages if the user inputs are incorrect.	M	Display Input field errors
FR4	Registered users shall be able to login into the system.	M	Login
FR5	The system shall be able to verify the user credentials	M	Verify user credentials
FR6	The system shall be able to display an error message if the user credentials are incorrect.	M	Display Login error
FR7	Registered users shall be able to select different statistical forecasting models to get the predictions from those models.	M	Select different statistical forecasting models
FR8	Registered users shall be able to select a time range for making predictions based on the number of months they want to forecast.	M	Select time range
FR9	Registered users shall be able to get the predictions of the tourist arrivals based on the inputs selected.	M	Get predicted tourist arrivals
FR10	Registered users shall be able to view the predicted tourist arrivals reports.	M	View predicted tourist arrivals reports
FR11	Registered users shall be able to download the predicted tourist arrivals reports in both excel and pdf format.	S	Download predicted tourist arrivals reports

FR12	Registered users shall be able to customize the predicted tourist arrivals reports.	S	Customize predicted tourist arrivals reports
FR13	Registered users shall be able to download the customized predicted tourist arrivals reports in both excel and pdf format.	S	Download customized predicted tourist arrivals reports
FR14	SLTDA shall be able to adjust the predictions based on the manual trend.	W	Adjust predictions
FR15	Registered users shall be able to view past and predicted tourist arrivals on two separate dashboards to make decisions quickly.	S	View dashboards
FR16	Registered users shall be able to customize the dashboards.	S	Customize dashboards
FR17	Registered users should be able to provide comments and suggestions.	C	Make suggestions
FR18	Registered users should be able to receive notifications	C	Receive notifications
FR19	Registered users shall be able to view customized past tourist arrivals reports.	S	View customized past tourist arrivals reports
FR20	Registered users shall be able to download the customized past tourist arrivals reports in both excel and pdf format.	S	Download customized past tourist arrivals reports

Table 4.2.1: Final list of functional requirements

4.2.2 Non-functional Requirements

Finalized Functional requirements have been presented in the following **Table 4.2.2**

Req ID	Constraints	Description
NFR1	Accuracy	The system should provide high-accuracy predictions since this system's data will be used for decision-making purposes.
NFR2	Response Time	The system should load within five seconds.
NFR3	Usability	The web interfaces should be simple and user-friendly to understand the features and functionalities of the system.
NFR4	Security	The security of the system should be at an acceptable level since the system can only be used by the authenticated user.
NFR5	Scalability	The application should be scalable to customize features and adapt to new technologies.
NFR6	Maintainability	The application should have its own self-service user manual to cover all features, so any user can read and learn its features and make the most of them.
NFR7	Usability	The application should have a contrast colour scheme to improve the usability of the system.
NFR8	Portability	The application should be responsive since uniformity in all the features and interface is to be active on both desktops and mobiles.
NFR9	Availability	The data should be available anytime for access simultaneously without any issue.

Table 4.2.2: Final list of non-functional requirements

At the initial stage, it was decided to use a simple colour schema as the theme of the designed prototype. However, according to the feedback collected by the first evaluator, it was decided to use a colourful theme rather than a simple colour schema for the prototype. Therefore, the non-functional requirement ID, NFR7, was different from the defined PSDP and the initial design.

4.3 Designed Implementation

4.3.1 Design Goals

The design of the prototype was implemented while keeping the design goals in mind in order to properly design the prototype. The proposed system aims to predict the number of tourist arrivals after the covid 19 outbreak while automating the manual tourist arrivals forecasting process in Sri Lanka. This would allow the Sri Lanka Tourism Development Authority and hotel managers to get an accurate prediction of the predicted number of tourist arrivals for the upcoming months. The system will not be considered correct if it is unable to accurately predict tourist arrivals. On the other hand, if the target audience finds the system difficult to use, it will not be regarded as reliable. Because of these facts, the system was designed with the goal of being correct and designing simple and user-friendly user interfaces to understand the features and functionalities of the system easily. This would result in improving the accuracy and usability of the designed prototype.

4.3.2 Colours in UI Design

Colour is the most simple and effective way of engaging users with the product in design implementation (Tapaniya, 2019). And also, these colours, as well as the contrast, can have a huge impact on the user accessibility of the product (Vaniukov, 2020). *"According to Joe Hallock's study, there is a significant difference in the choice of colour between genders"*, so that in terms of gender, there is a high possibility of using the proposed system by males rather than females. This was identified during the interview process. The research by Natalia Khouw on "The Meaning of Colour for Gender" found that men like gray, white, or black better than women (Vaniukov, 2020). As a result, it was decided to design this proposed web-based tourism demand forecasting system with a simple colour scheme of grey, white, and black in the initial design stage **(Figure 4.3.1)**.

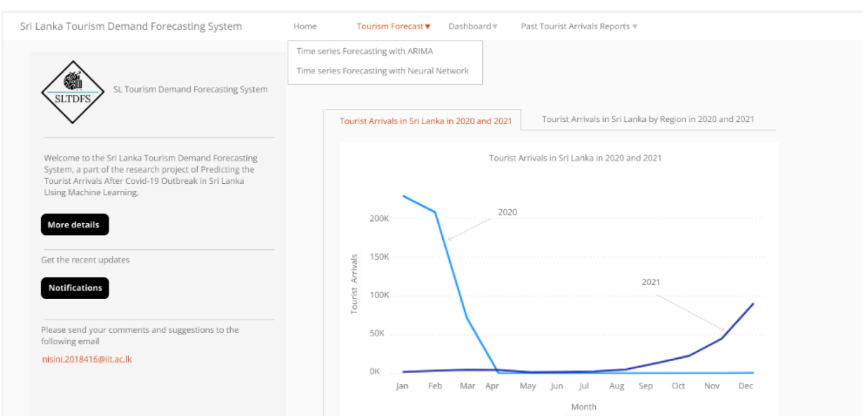


Figure 4.3.1: Initial design of the prototype

However, after collecting the feedback from the first evaluators (refer to **User Testing**) for the developed prototype, most of them preferred to have a contrast colour theme for the website, so it was decided to change the colour of the website's theme and develop it again. As a result, the green colour was chosen to be used as the theme of the website because it is one of the most contrasting colours (Alscher, 2019). On the other hand, since the green colour brings calm feelings of renewal (Vaniukov, 2020), it would be most appropriate for the tourism industry.

4.3.3 Dashboards Design

In terms of dashboard design, two dashboards were created to display the past and predicted tourist arrivals separately. These dashboards were designed by considering the perspective of the target audience so that they could have a better understanding of the tourist arrival past data as well as the forecast data that was calculated from the ARIMA and NNAR models. At the initial stage, the past tourist arrivals dashboard was designed using only the past tourist arrivals in 2020 and 2021 **(Figure 4.3.2)**.

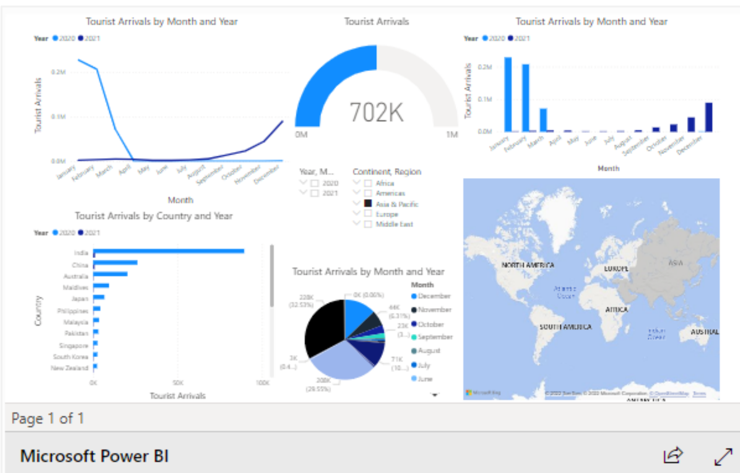


Figure 4.3.2: Designed Dashboard for Past Tourist Arrivals at the initial stage

However, with the feedback from the first evaluators, it was decided to redesign the past tourist arrivals dashboard, including seven different pages for the separate year ranges from 1997 to 2021. On the last page, all the past tourist arrivals from 1977 to 2021 have been included in the redesigned past tourist arrivals dashboard **(Figure 4.3.3)**.

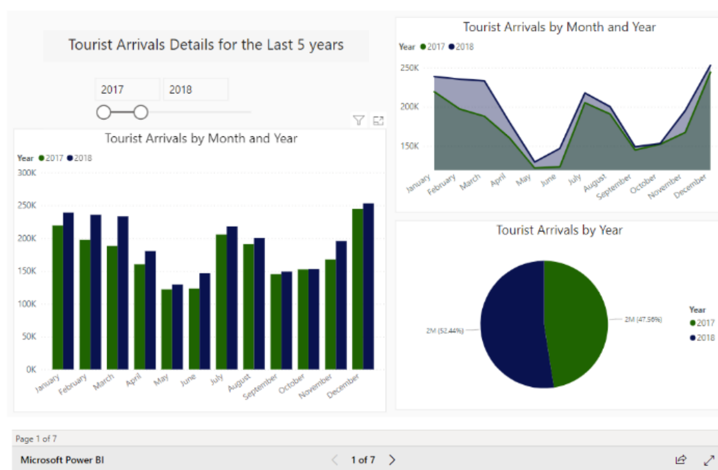


Figure 4.3.3: Redesigned Dashboard for Past Tourist Arrivals after the feedback of the first evaluators

Figure 4.3.4: Designed Dashboard for Predicted Tourist Arrivals at the initial stage

The predicted tourist arrivals dashboard was initially designed by focusing on a single-page dashboard that included all of the forecasts produced using the ARIMA and NNAR models. In addition, the comparison prediction results of these two models have been displayed on that same page **(Figure 4.3.4)**.

However, as per the feedback from the first evaluators, it was decided to modify the predicted dashboard also. As a result, three pages were created separately for ARIMA forecasts, NNAR forecasts, and the comparative predictions of these two models in this redesigned dashboard **(Figure 4.3.5)**.

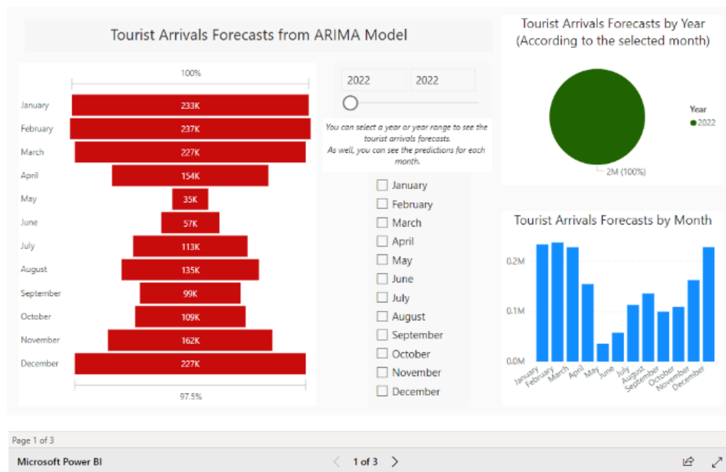


Figure 4.3.5: Redesigned Dashboard for Predicted Tourist Arrivals after the feedback of the first evaluators

4.4 Design Tools

4.4.1 Prototype Design Tool

Among the prototype tools such as Pencil, Figma, and Justinmind, **Figma** was chosen over the other tools because of its ease of use. And also, it is one of the most appropriate tools for designing web-based prototype applications.

4.4.2 User Stories

User stories provide an essential foundation for the design stage. They are useful in identifying the exact user goals and can also be used as a way to organize and prioritize designing each user interface (VM@CanvasFlip, 2018). As a result, the following user stories **(Table 4.4.1)** were created in the user interface design process.

User story #	User story title	User role	Feature	Goal	Priority	Effort
S1	Register with the system	As an unregistered user	I want to be able to create an account	So that I can register with the system	M	400
S2	Login	As a registered user	I want to be able to login into the system	So that I can get the predictions of tourist arrivals	M	400
S3	Select different statistical forecasting models	As a registered user	I want to be able to select different statistical forecasting models	So that I can get the tourist arrivals predictions from these different statistical models	M	400
S4	Select a time range	As a registered user	I want to be able to select a time range for making predictions based on the number of months that I want to forecast	So that I can get the predicted tourist arrivals based on the number of months that I want to forecast	M	400
S5	Get the predicted tourist arrivals	As a registered user	I want to be able to get the predictions of the tourist arrivals based on the inputs that I selected	So that I can get the predicted number of tourist arrivals based on my input selections	M	400

S6	View predicted tourist arrivals reports	As a registered user	I want to be able to view the predicted tourist arrivals reports	So that I can use those data to manage my business operations	M	400
S7	Download predicted tourist arrivals reports	As a registered user	I want to be able to download the predicted tourist arrivals reports in both excel and pdf format	So that I can use those downloaded data to make my own reports	S	300
S8	Customize predicted tourist arrivals reports	As a registered user	I want to be able to customize the predicted tourist arrivals reports.	So that I can view the predicted tourist arrivals reports according to my requirements	S	300
S9	Download customized predicted tourist arrivals reports	As a registered user	I want to be able to download the customized predicted tourist arrivals reports in both excel and pdf format	So that I can use those downloaded customized predicted tourist arrivals details according to my requirements	S	300
S10	Adjust predictions	As a SLTDA officer	I want to be able to adjust the predictions based on the manual trend	So that I can perform the current manual trend analysis using this system	W	100
S11	View dashboards	As a registered user	I want to be able to view past and predicted tourist arrivals on two separate dashboards	So that I can make decisions quickly	S	300

S12	Customize dashboards	As a registered user	I want to be able to customize the dashboards	So that I can get insights from these dashboards according to my requirements	S	300
S13	Make comments and suggestions	As a registered user	I want to be able to provide comments and suggestions to improve the system	So that I can get the maximum benefit from this system	C	200
S14	Receive notifications	As a registered user	I want to be able to receive notifications	So that I can keep in touch with the latest updates	C	200
S15	View customized past tourist arrivals reports	As a registered user	I want to be able to view customized past tourist arrivals reports	So that I can view the past tourist arrivals reports according to my requirements	S	300
S16	Download customized past tourist arrivals reports	As a registered user	I want to be able to download the customized past tourist arrivals reports in both excel and pdf format.	So that I can easily compare the predicted tourist arrivals report with these past tourist arrivals reports	S	300
S17	Get accurate predictions	As a registered user	I want to be able to get high-accuracy predictions from the system to do the operations	So that I can manage the operations more efficiently	M	400

S18	Response Quickly	As a registered user	I want to be able to get predicted data from the system within five seconds	So that I can get the predictions without waiting time	S	300
S19	Usability of the web interfaces	As a registered user	I want to be able to have a contrast colour scheme and user-friendly web interfaces	So that I can easily understand the features and functionalities of the system	S	300
S20	Security	As a registered user	I want to be able to get at an acceptable level of security	So that I can secure my data	S	300
S21	Customize features	As a registered user	I want to be able to customize features and adapt the application to new technologies	So that I can fulfill my expectations according to my requirements.	C	200
S22	Self-service user manual	As a registered user	I want to be able to have a self-service user manual to cover all features in the system	So that I can read and learn its features	C	200

Table 4.4.1: User Stories

Please refer to **Appendix C – Class Diagram**

5. METHODOLOGY

5.1 Selected Development Approach

Among the software development approaches of structured and object-oriented approaches, the **object-oriented approach** was chosen. The reason for that is, in the object-oriented approach, the focus is on encapsulating the structure and behavior of information systems into discrete modules that connect data and process, with the aim of improving the quality and productivity of system analysis and design by making it more useful. It is the best approach for this project since it is suitable for most business applications that are expected to be customized or extended, as well as object-oriented design programming done concurrently with other phases.

5.2 Selected Methodology

This research study has chosen **agile** as the project management/ software development methodology among the agile and traditional approaches. This is because, according to the nature of the project, the scope and requirements can change at any time during the project. Therefore, being agile will be the best option for meeting all of the constantly changing needs. And also, the user stories that are specific to agile were used in the design stage of the project. On the other hand, this research will be used for **both quantitative data**, such as the numerical data of tourist arrivals and performance evaluation values, **and qualitative data**, which will be gathered from the interview responses. Therefore, the **mix approach**, which is a combination of both quantitative and qualitative methods was chosen as the appropriate primary data collection methodology for this project.

5.3 Hybrid Methodology

Although agile was chosen as the suitable methodology, the planning strategies can be combined with both agile and traditional methods. In such a situation, the hybrid methodology can be used to successfully complete the project. For instance, the detailed activity schedules, which are related to the **traditional approach methodology**, have been used in order to complete the project tasks sequentially.

6. TOOLS AND IMPLEMENTATION

6.1 Tools and Skills

6.1.1 Tools

The following **Table 6.1.1** presents the tools that were used to implement the proposed web-based tourism demand forecasting system.

Tool	Justification
Programming Environments & Languages	
R programming language	R is an open-source statistical programming language that can be used in many areas, such as machine learning, statistical algorithms, and data science (Study.com, 2022). As a result, **R** was chosen as the programming language because the research project is directly related to statistical data analysis and machine learning.
R Studio	R Studio is an IDE (Integrated Development Environment) for the R programming language. Since R was selected as the programming language, **R Studio** has been chosen as the programming environment.
Libraries (R packages)	
caret, forecast, tseries, zoo	Caret, forecast, tseries, and zoo are the R packages that were chosen to build the machine learning models such as ARIMA and NNAR. It is mainly because the ARIMA model was built with the **caret** and **tseries** libraries, whereas the NNAR model was built with the **zoo** library. The **forecast** package was used to make the predictions of the tourist arrivals from these two models.
shiny	It was decided to build the proposed web application with the R programming language, and since shiny is an R package that makes it easy to build interactive web apps, **shiny** was chosen to implement the proposed web application.

shinythemes	The shinythemes is also an R package that includes several Bootstrap themes that can be used with Shiny applications. Since shiny was decided to implement the proposed web app, the **shinythemes** package was chosen to decorate the web app.
shinyalert	The **shinyalert** package, which comes under the R package, is capable of creating pretty popup messages in Shiny, and it was decided to use this package to display error messages when validating the user inputs during the user registration process.
shinyauthr	The **shinyauthr** package was chosen to be used in the login process in order to add in-app user authentication to the Shiny Web App.
shinycssloaders	In order to show loader animations automatically while a Shiny output is (re)calculating, the **shinycssloaders** package was used.
RMySQL	The **RMySQL** package was chosen since MySQL was selected as the database for this project, and RMySQL is a database interface and MySQL driver for R.
DBI	To connect R to the database management system (DBMS), the **DBI** package was chosen because it helps to separate the connectivity to the DBMS into a "front-end" and a "back-end".
tidyverse	The **tidyverse** package was chosen because it is a group of packages that work together since they share data representations and an "API" architecture. I used the package dplyr which is one of the tidyverse packages for data manipulation tasks.
plotly	To create interactive web-based graphs of past tourist arrivals, the **plotly** package was chosen.
highcharter	**Highcharter** package includes Highcharts javascript library and its modules. It was also used in the creation of predicted tourist arrival graphs.
lubridate	This **lubridate** package was chosen because it provides the ability to incorporate with date-time, which is important for this project.
DT	The **DT** package was chosen because it provides a R interface to the JavaScript library DataTables, which was utilized in the web-based

	tourism demand forecasting system to display past and predicted tourist arrivals data.
Database	
MySQL	**MySQL** was chosen as the database because the entire data of the project is structured data, and MySQL is a type of structured database that only stores structured data.
Data Visualization	
Power BI	Because the project involves a small amount of data, **Power BI** was chosen over the other options. It's because when the amount of data is limited, Power BI performs better and faster.
Deployment Platform	
shinyapps.io	Because the web application was built with Shiny, it was chosen to be deployed using **shinyapps.io**. It is a free and simple platform to share Shiny applications.

Table 6.1.1: Tools

6.1.2 Skills

There are numerous existing skills that I have developed during the project development. During the final year, we learned about the R statistical programming language for the Business Intelligence module. With the knowledge that I have gained, I was able to extend my knowledge of the R programming language since my whole project was developed using the R statistical programming language. Before implementing my web-based tourism demand forecasting, I had only knowledge about how to build machine learning algorithms such as K Means Clustering, Neural Networks, K Nearest Neighbor, Support Vector Machines, Decision Trees, and so on. However, during the development process, I did quite a bit of research on time series analysis models, including the ARIMA model and all that. It was a huge investment for me to improve my knowledge of machine learning algorithms as well as the R statistical programming language. In addition, I self-studied and learned how to develop R Shiny web application which is utilized in my project. It was a new skill that I improved by myself for the project. Furthermore, I have learned how to create dashboards using Power BI by myself. It also helped me develop a new skill on the data visualization side.

6.2 Implementation

6.2.1 Key Functionality Implementation

Database Connection

The most challenging part of an R Shiny web application is to maintain the connection with the database (Appsilon | Enterprise R Shiny Dashboards, 2022). The first step in creating the connection to the database is to create a connection to a remote MySQL database, which was done by using dbConnect. The R packages of RMySQL and DBI were also used to make the database connection.

```
#to make the connection to the database
sltdfs_db <- dbConnect(MySQL(),
                       dbname='sltdfs',
                       user='root',
                       password='',
                       host='localhost',
                       port=3306)

#write data into database
write_query_fun <- function(name, df){
  sltdfs_db <- dbConnect(MySQL(),
                         dbname='sltdfs',
                         user='root',
                         password='',
                         host='localhost',
                         port=3306)

  dbWriteTable(conn = sltdfs_db, name = name, value = df, overwrite = FALSE, append = TRUE,
               row.names = FALSE)


}
```

In the first argument, the MySQL function is the driver that I used to connect to the MySQL database. Next, I have given the name of the database along with its credentials to authenticate myself. Finally, I have given the database host and the port that I want to connect to. The result of DB Connect is a DBIConnection object, which I have used when writing the user details into the database in the registration process. The database connection and the write query functions have been created out of the server function, while the validation process of the registration has been included within the server function in the server.R file. The user interface for the registration process has been developed in the ui.R file.

```
register_tab <- tabPanel(
  title = icon("user"),
  value = "register",
  shinyauthr::loginUI("login"),
  fluidRow(
    column(width = 4, offset = 4,
           h2("Please Register"),
           textInput("first_name", "First Name:", placeholder = "Enter your first name"),
           textInput("last_name", "Last Name:", placeholder = "Enter your last name"),
           textInput("email", "Email:", placeholder = "Enter your email"),
           textInput("contact_number", "Contact Number", placeholder = "Enter your contact number"),
           textInput("user", "User Name", placeholder = "Enter your user name"),
           passwordInput("password", "Password", placeholder = "Enter a password"),
           passwordInput("con_password", "Confirm Password", placeholder = "Enter the same password"),
    ),
    column(width = 3, offset = 5,
           actionButton("registration_button","Register"),
           ),
  ),
)
```

```
  observeEvent(input$registration_button, {
    if (input$first_name == ""){
      shinyalert("You need to enter your first name!", type = "error")
    } else if (input$last_name == ""){
      shinyalert("You need to enter your last name!", type = "error")
    } else if (input$email == ""){
      shinyalert("You need to enter your email!", type = "error")
    } else if (input$contact_number == ""){
      shinyalert("You need to enter your contact number!", type = "error")
    } else if (input$user == ""){
      shinyalert("You need to enter your username!", type = "error")
    } else if (input$password == ""){
      shinyalert("You need to enter a password!", type = "error")
    } else if (input$con_password == "" | input$con_password != input$password){
      shinyalert("You need to enter the same password!", type = "error")
    }else {
      df <- data.frame(
        "first_name" = input$first_name,
        "last_name" = input$last_name,
        "email" = input$email,
        "contact_number" = input$contact_number,
        "user"= input$user,
        "password" = input$password,
        "created_at" = Sys.time()
      )

      shinyalert("You have registered successfully!", type = "success")
      session$reload()
    }

  })
}
```

User authentication

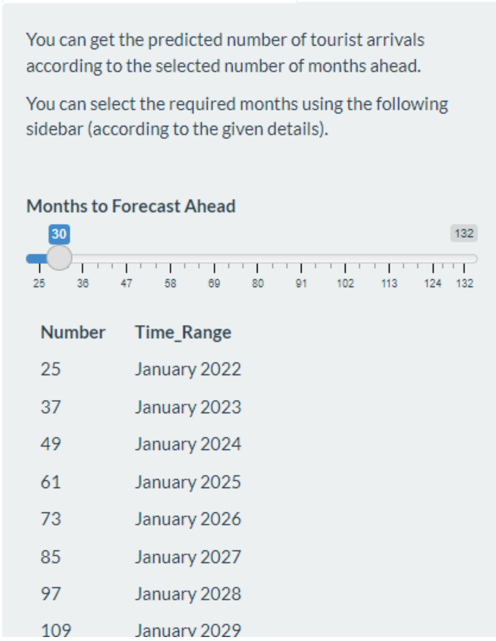
User authentication is also one of the key and challenging parts since it connects to the database. Here, I have used the shinyauthr package to create the login option. The user interface has been developed within the ui.R file. And all the database functions and code for validations have been included outside of the server function which is in the server.R file.

```
# login tab ui to be rendered on launch
login_tab <- tabPanel(
  title = icon("lock"),
  value = "login",
  shinyauthr::loginUI("login")
)
```

Here, I have created a reference to the users table and the dbConnection object and stored it in the new object. After that, the user data will be retrieved using the pipe operator (%>%) in the dplyr methodology when the user tries to login to the system. This operator has been used in the data retrieval from the loaded csv files also.

```
#to make a reference to users
sltdfs_database <- tbl(sltdfs_db, "users")

user_credentials <- sltdfs_database %>%
  select(user, password) %>%
  collect()

user_credentials_df <- data.frame(user_credentials)

# user database for logins
user_base <- tibble::tibble(
  user = c(user_credentials_df$user),
  password = purrr::map_chr(c(user_credentials_df$password), sodium::password_store)
)
```

Side bar Implementation



The sidebar is the most important part of this website because it fulfills the function that allows users to select the required number of months to forecast ahead in order to get the tourist arrivals predictions for up to the selected number of months. Numbers are categorized by considering the number of months to get the predictions for the upcoming years. Since the past tourist arrivals data from January 2020 to December 2021 was used as the testing data, a minimum of 24 months had to be kept for the testing data that was used to build the model. Therefore, it was decided to start the sidebar with the number 25, which refers to January 2022, and divide the sidebar by twelve in order to keep the space between each categorized number. Since the predictions were calculated up to December 2030, which means 132 months, the sidebar was categorized from 25 to 132. To make it easy to understand that categorization, each divided number was introduced with its month and year.

```
#-----------------------number categorization for sidebar------------------------

Number <- c("25", "37", "49", "61", "73", "85", "97", "109", "121")
Time_Range <- c("January 2022","January 2023","January 2024","January 2025",
                "January 2026","January 2027","January 2028","January 2029","January 2030")
month_df <- data.frame(Number, Time_Range)
```

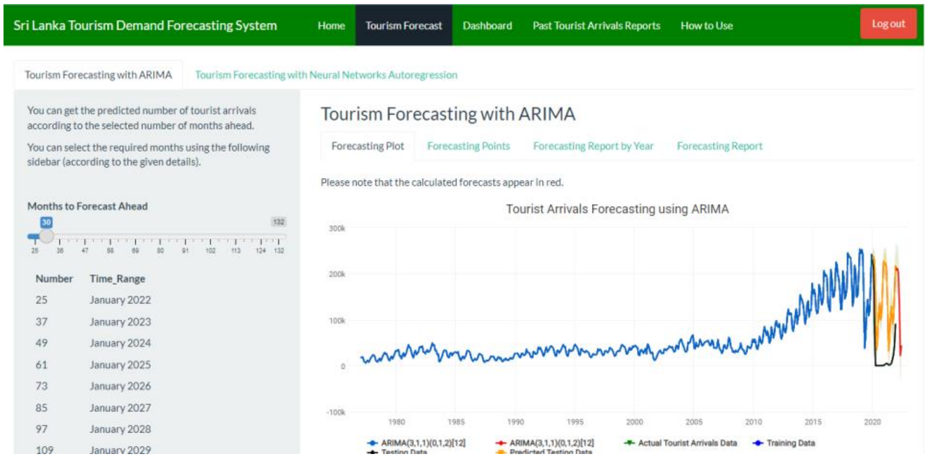
Two sidebars have been developed to select the required number of months to forecast ahead from each model.

```
#------------------------tourism_forecast_tab------------------------------------

tourism_forecast_tab <- tabPanel(
  title = "Tourism Forecast",
  value = "tourism_forecast",
  tabsetPanel(
    tabPanel("Tourism Forecasting with ARIMA",
             sidebarLayout(
               sidebarPanel(
                 p("You can get the predicted number of tourist arrivals according to the selected number of months ahead."),
                 p ("You can select the required months using the following sidebar (according to the given details)."),
                 hr(),

                 sliderInput(
                   inputId = "forecast_n_months",
                   label = strong("Months to Forecast Ahead"),
                   min = 25,
                   max = 132,
                   value = 30),

                 tableOutput("month_view_1"),

               ),
```

```
tabPanel("Tourism Forecasting with Neural Networks Autoregression",
         sidebarLayout(
           sidebarPanel(
             p("You can get the predicted number of tourist arrivals according to the selected number of months ahead."),
             p ("You can select the required months using the following sidebar (according to the given details)."),
             hr(),
             sliderInput(
               inputId = "forecast_n_months_2",
               label = strong("Months to Forecast Ahead"),
               min = 25,
               max = 132,
               value = 40),

             tableOutput("month_view_2"),
           ),
```

Displaying ARIMA forecast plot based on the sidebar selection

Users will be able to get the predictions for the upcoming months that they want to forecast by selecting the required number of months from the sidebar. Then the ARIMA plot will be automatically generated for up to the selected number of months accordingly. The ARIMA model



was built by using R packages such as caret, forecast, and tseries. The past tourist arrivals statistics from January 1977 to December 2021 were used as the dataset. After loading the required libraries and the data set, the data preprocessing task of NA values was checked. The data from January 1977 to December 2019 was considered the training data, whereas the rest of the data was considered the testing data. Therefore, the data was split into training and testing accordingly. After that, the "Tourist Arrival" column of the training dataset was translated into time series format using the ts function. Here, the frequency was specified as 12 because the predictions would be calculated using the monthly data. Then the ARIMA model was built using the auto.arima function. When it comes to getting the predictions using the ARIMA model, here, the forecasting points for up to two years (2*12) were calculated in order to validate the testing data because this testing data consists of the years 2020 and 2021. After that, the predictions were calculated up to the year 2030 by defining h = 9*12.

```
#-----------------pre-processing task------------------------------

#check NA values in arrivals
any(is.na(arrivals))


#------------splitting arrivals data frame into training and testing-------------

arrivals_train <- arrivals[1:516, ]
arrivals_test <- arrivals[517:540, ]

#convert to time series format
arrivals_timeseries <- ts(arrivals_train$Tourist_Arrivals, start = c(1977,1),
                          end = c(2019,12), frequency = 12)


#----------------------------ARIMA model-----------------------------------------

arima_model <- auto.arima(arrivals_timeseries)
summary(arima_model)

#testing the predicted data
test_forecast_arima <- forecast(arima_model, h=2*12)

#predictions up to 2030
forecast_arima <- forecast(arima_model, h=9*12)
```

Then the Highcharter was used to display the predictions of the ARIMA model using the built model and the user input of number of months to forecast ahead.

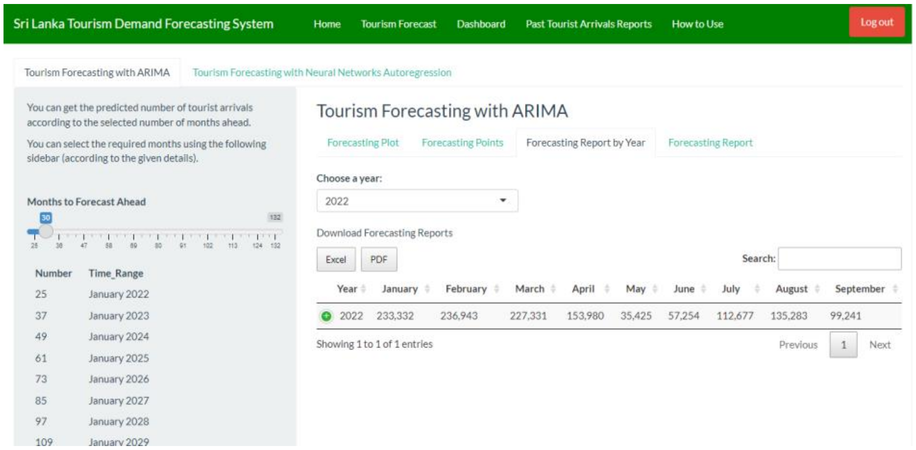
```
mainPanel(
  h3("Tourism Forecasting with ARIMA"),
  tabsetPanel(
    tabPanel("Forecasting Plot",
             br(),
             p("Please note that the calculated forecasts appear in red."),
             withSpinner(highchartOutput("forecastPlot"), type = 4),
             textOutput("summary_arima"),
    ),
```

The "forecastPlot" parameter of the highchartOutput function was defined as follows. Apart from the two series generated by the ARIMA plot, four additional series were added to the plot to help the user better understand the generated forecasts. These four additional series represent actual tourist arrival details from January 1977 to December 2021, the training and testing data that were used to build the model, as well as predictions that were calculated for testing data respectively.

```
output$forecastPlot <- renderHighchart({
  forecast_arima_plot <- forecast(arima_model,
                                  input$forecast_n_months) %>%
    hchart %>%
    hc_add_series(ts(arrivals$Tourist_Arrivals, start = c(1977,1),
                     end = c(2021,12), frequency = 12), color = "green",
                  name = "Actual Tourist Arrivals Data") %>%
    hc_add_series(ts(arrivals_train$Tourist_Arrivals, start = c(1977,1),
                     end = c(2019,12), frequency = 12), color = "blue",
                  name = "Training Data") %>%
    hc_add_series(ts(arrivals_test$Tourist_Arrivals, start = c(2020,1),
                     end = c(2021,12), frequency = 12), color = "black",
                  name = "Testing Data") %>%
    hc_add_series(data = test_forecast_arima, color = "orange",
                  name = "Predicted Testing Data") %>%
    hc_add_theme(hc_theme_google()) %>%
    hc_title(text = "Tourist Arrivals Forecasting using ARIMA")
})

    output$summary_arima <- renderPrint({
      summary(arima_model)
    })
```

Displaying ARIMA forecasting reports based on the year selection

Under the "Forecasting Report by Year" tab, users will be able to view the ARIMA forecasting



reports based on the selected year that they want to view the predictions for. In addition, they will also be able to download these generated reports in both excel and pdf format. This section has been implemented as follows,

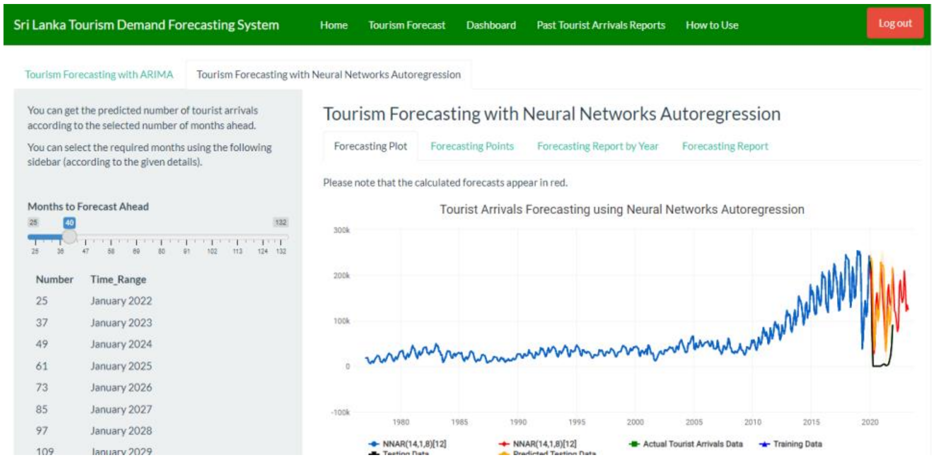
```
tabPanel("Forecasting Report by Year",
         br(),
         selectInput("select_year_with_updateSelectInput_arima",
                     label = "Choose a year:",
                     choices = NULL),
         p("Download Forecasting Reports"),
         withSpinner(DTOutput("report_arima"), type = 4),
),
```

The year selection option and the report downloading option were developed as follows,

```
updateSelectInput(session,
                  "select_year_with_updateSelectInput_arima",
                  choices = unique(arima_forecast$Year))


output$report_arima <- renderDT({

  arima_forecast %>%
    filter(Year == input$select_year_with_updateSelectInput_arima) %>%
    datatable(rownames = FALSE,
              extensions = c('Responsive', 'Buttons'), options = list(
                orientation ='landscape',
                dom = 'Bfrtip',
                buttons =
                  list ('excel', list(
                    extend = 'pdf',
                    pageSize = 'A4',
                    orientation = 'landscape',
                    filename = 'ARIMA_Forecasting_Report_by_Year'
                  )
                  )
              )
    )

})
```

Displaying NNAR forecast plot based on the user selection

Users will also be able to get predictions using the NNAR model, and all the functions that have been developed to display the ARIMA plot were used to display the NNAR plot based on the user input. The dataset and the same ratio of training and testing data that was used to build the ARIMA model were used to build the NNAR model.




```
#--------------------------Neural Network AR model-------------------------------

set.seed(124)

nnar_model = nnetar(arrivals_timeseries)
summary(nnar_model)

#testing the predicted data
test_forecast_nnar = forecast(nnar_model, h =2*12)

#predictions up to 2030
forecast_nnar <- forecast(nnar_model, h=9*12)
```

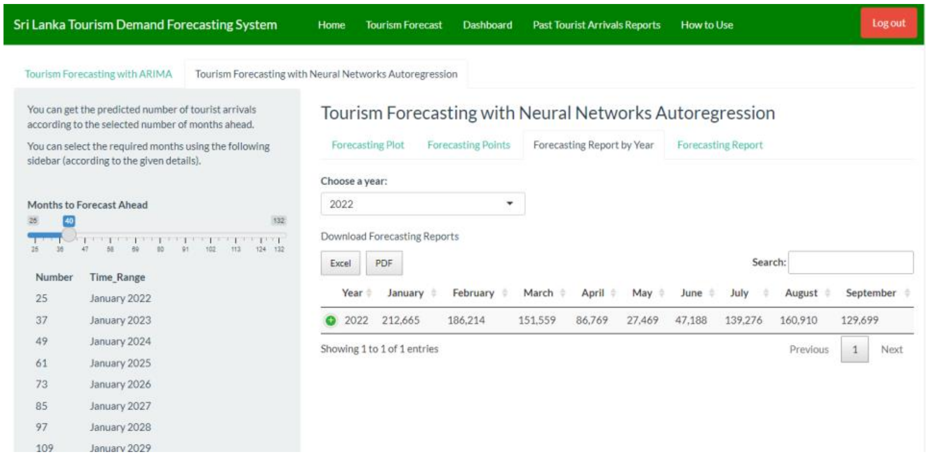
To display the NNAR plot, the Highcharter was used here also.

```
mainPanel(
  h3("Tourism Forecasting with Neural Networks Autoregression"),
  tabsetPanel(
    tabPanel("Forecasting Plot",
             br(),
             p("Please note that the calculated forecasts appear in red."),
             withSpinner(highchartOutput("tn_forecastPlot"), type = 4),
             textOutput("summary_nnar"),
    ),
```

The tn_forecastPlot parameter passing through the highchartOutput function is defined as follows. The same additional four series that were added to the ARIMA model were also added to the NNAR plot.

```
output$tn_forecastPlot <- renderHighchart2({
  forecast_nnar_plot <- forecast(nnar_model,
                               input$forecast_n_months_2) %>%
    hchart %>%
    hc_add_series(ts(arrivals$Tourist_Arrivals, start = c(1977,1),
                    end = c(2021,12), frequency = 12), color = "green",
                 name = "Actual Tourist Arrivals Data") %>%
    hc_add_series(ts(arrivals_train$Tourist_Arrivals, start = c(1977,1),
                    end = c(2019,12), frequency = 12), color = "blue",
                 name = "Training Data") %>%
    hc_add_series(ts(arrivals_test$Tourist_Arrivals, start = c(2020,1),
                    end = c(2021,12), frequency = 12), color = "black",
                 name = "Testing Data") %>%
    hc_add_series(data = test_forecast_arima, color = "orange",
                 name = "Predicted Testing Data") %>%
    hc_add_theme(hc_theme_google()) %>%

    hc_title(text = "Tourist Arrivals Forecasting using Neural Networks Autoregression")
})
```

Displaying NNAR forecasting reports based on the year selection



The same functions that were used to generate ARIMA forecasting reports were utilized to generate NNAR forecasting reports based on user input. The following is how this section was implemented.

```
tabPanel("Forecasting Report by Year",
         br(),
         selectInput("select_year_with_updateSelectInput_nnar",
                     label = "Choose a year:",
                     choices = NULL),
         p("Download Forecasting Reports"),
         withSpinner(DTOutput("report_nnar"), type = 4),
),
```

The year selection option and the report downloading option were developed as follows,

```
updateSelectInput(session,
                  "select_year_with_updateSelectInput_nnar",
                  choices = unique(nnar_forecast$Year))

output$report_nnar <- renderDT({

  nnar_forecast %>%
    filter(Year == input$select_year_with_updateSelectInput_nnar) %>%
    datatable(rownames = FALSE,
              extensions = c('Responsive', 'Buttons'), options = list(
                orientation ='landscape',
                dom = 'Bfrtip',
                buttons =
                  list ('excel', list(
                    extend = 'pdf',
                    pageSize = 'A4',
                    orientation = 'landscape',
                    filename = 'NNAR_Forecasting_Report_by_Year'
                  )
                  )
              )
    )
})
```

Displaying past tourist arrivals reports

There are two kinds of past tourist arrival reports that allow users to view details of past tourist arrivals. These reports were implemented as follows,

```
#---------------------------past_reports_tab--------------------------------------

past_reports_tab <- tabPanel(
  title = "Past Tourist Arrivals Reports",
  value = "past_report",
  tabsetPanel(
```

```
    tabPanel("Tourist Arrivals by Year and Month",
             sidebarLayout(
               sidebarPanel(
                 selectInput("select_year_with_updateSelectInput_1",
                             label = "Choose a year:",
                             choices = NULL),

                 downloadButton("download_data_1", "Download Full Data", class="btn btn-primary"),
               ),
               mainPanel(
                 h3("Tourist Arrivals by Year and Month"),
                 p("Download Past Reports"),
                 withSpinner(DTOutput("report_1"), type = 4),
               ),
             ),
    ),
    tabPanel("Tourist Arrivals by Continent, Region and Country",
             sidebarLayout(
               sidebarPanel(
                 selectInput("select_year_with_updateSelectInput_2",
                             label = "Choose a year:",
                             choices = NULL),
                 selectInput("selected_continent",
                             label = "Choose a continent:",
                             choices = NULL),
                 selectInput("selected_region",
                             label = "Choose a region:",
                             choices = NULL),
                 selectInput("selected_country",
                             label = "Choose a country:",
                             choices = NULL),
                 downloadButton("download_data_2", "Download Full Data", class="btn btn-primary"),
               ),
               mainPanel(
                 h3("Tourist Arrivals by Continent, Region and Country"),
                 p("Download Past Reports"),
                 withSpinner(DTOutput("report_2"), type = 4),
               ),
             ),
    ),
  ),
)
```

In the "Tourist Arrivals by Year and Month" report, the year selection function was implemented as follows,

```
updateSelectInput(session,
                  "select_year_with_updateSelectInput_1",
                  choices = unique(arrivals_details$Year))
```

The data retrieval based on the year selection as well as the downloading options were implemented as follows,

```
output$report_1 <- renderDT({

  arrivals_details %>%
    arrange(desc(Year)) %>%
    filter(Year == input$select_year_with_updateSelectInput_1) %>%
    datatable(rownames = FALSE,
              extensions = c('Responsive', 'Buttons'), options = list(
                pageLength = 12,
                orientation ='landscape',
                dom = 'Bfrtip',
                buttons =
                  list('excel', list(
                    extend = 'pdf',
                    pageSize = 'A4',
                    orientation = 'landscape',
                    filename = 'Tourist_Arrivals_by_Year_and_Month'
                  ))
              )
    )

})

output$download_data_1 <- downloadHandler(
  filename = "tourist-arrivals-by-year-and-month.csv",
  content = function(file){
    file.copy("tourist_arrivals_details.csv",
              file)
  }
)
```

```
output$report_1 <- renderDT({

  arrivals_details %>%
    arrange(desc(Year)) %>%
    filter(Year == input$select_year_with_updateSelectInput_1) %>%
    datatable(rownames = FALSE,
              extensions = c('Responsive', 'Buttons'), options = list(
                pageLength = 12,
                orientation ='landscape',
                dom = 'Bfrtip',
                buttons =
                  list('excel', list(
                    extend = 'pdf',
                    pageSize = 'A4',
                    orientation = 'landscape',
                    filename = 'Tourist_Arrivals_by_Year_and_Month'
                  ))
              )
    )

})
```

In the "Tourist Arrivals by Continent, Region, and Country" report, the functions for the input selections (year, continent, region, and country) were implemented as follows,

```
updateSelectInput(session,
                  "select_year_with_updateSelectInput_2",
                  choices = unique(cou_arrivals$Year))

updateSelectInput(session,
                  "selected_region",
                  choices = unique(cou_arrivals$Region))

observeEvent(c(input$selected_region),
             {
               countries_in_regions <- cou_arrivals %>%
                 filter(Region == input$selected_region) %>%
                 pull(Country)

               updateSelectInput(session,
                                 "selected_country",
                                 choices = countries_in_regions)
             })

updateSelectInput(session,
                  "selected_continent",
                  choices = unique(cou_arrivals$Continent))

observeEvent(c(input$selected_continent),
             {
               regions_in_continents <- cou_arrivals %>%
                 filter(Continent == input$selected_continent) %>%
                 pull(Region)

               updateSelectInput(session,
                                 "selected_region",
                                 choices = regions_in_continents)

             })
```

The data retrieval based on the user's selection as well as the downloading options were implemented as follows,

```
output$download_data_2 <- downloadHandler(
  filename = "tourist-arrivals-by-continent-and-region.csv",
  content = function(file){
    file.copy("country_by_arrivals.csv",
              file)
  }
)


output$report_2 <- renderDT({

  cou_arrivals %>%
    select(Year, Month, Continent, Region, Country, Tourist_Arrivals) %>%
    arrange(desc(Year)) %>%
    filter(Year == input$select_year_with_updateSelectInput_2) %>%
    filter(Region == input$selected_region) %>%
    filter(Country == input$selected_country) %>%
    group_by(Year, Month, Continent, Region) %>%
```

```
datatable(
  rownames = FALSE,
  extensions = c('Responsive', 'Buttons'), options = list(
    pageLength = 12,
    orientation ='landscape',
    dom = 'Bfrtip',
    buttons =
      list('excel', list(
        extend = 'pdf',
        pageSize = 'A4',
        orientation = 'landscape',
        filename = 'Tourist_Arrivals_by_Continent_Region_and_Country'
      ))
  )
)

})
```

Please refer to **Appendix D – Other Functionalities**

You can access the Sri Lanka Tourism Demand Forecasting System using the following link, https://sltdfs.shinyapps.io/Sri_Lanka_Tourism_Demand_Forecasting_System/ (Credentials – Username: user1, Password: pass1)

7. TESTING

7.1 Functional Testing

7.1.1 Black-box testing

The black box testing has been performed in the following **Table 7.1.1.**

#ID	Description	Sample Input	Expected Output	Actual Output	Status
1	Register	**First Name:** Sandamini **Last Name:** Wijesiriwardana **Email:** lekhasandamini@gmail.com **Contact Number:** 0771561504 **User Name:** Sandamini **Password:** sandamini123 **Confirm Password:** sandamini123	- Displaying a registration success message and load the login page - Updating the database	Same as expected	Pass

2	Register	**First Name:** Sandamini **Last Name:** **Email:** **Contact Number:** **User Name:** **Password:** sandamini123 **Confirm Password:**	Displaying an error message named "You need to enter your last name!". (Based on the first missing value, an error message should be displayed)	Same as expected	Pass
3	Login	**User Name:** Sandamini **Password:** sandamini123	Login success and load the home page	Same as expected	Pass
4	Login	**User Name:** Sandamini **Password:** sandamini	Displaying login error "Invalid username or password!"	Same as expected	Pass
Testing Tourism Forecasting with ARIMA					
Testing Tab - Forecasting Plot					
5	Getting predictions using the ARIMA model	Bring the sidebar near 36	Generating the ARIMA plot for up to December 2022	Same as expected	Pass
Testing Tab - Forecasting Report by Year					
6	View Forecasting Reports by Year	Choose the 2023 as the year	Generating tourist arrivals forecast report for the year 2023	Same as expected	Pass
7	View the tourist arrivals for the rest of the months which are not displayed in the page	Click on the expand icon	Expanding the report and displaying the rest of the months with the number of tourist arrivals	Same as expected	Pass
8	Downloading Forecasting Report in excel format	Click on the "Excel" option	Downloading the forecasting report for the selected year in excel format	Same as expected	Pass

9	Downloading Forecasting Report in pdf format	Click on the “Pdf” option	Downloading the forecasting report for the selected year in pdf format	Same as expected	Pass
Testing Tab - Forecasting Report					
10	Search details using search bar	Search on the year 2024	Displaying the tourist arrivals for each month of the year 2024	Same as expected	Pass
11	View the tourist arrivals for the rest of the months which are not displayed in the page	Click on the expand icon	Expanding the report and displaying the rest of the months with the number of tourist arrivals	Same as expected	Pass
12	Downloading Forecasting Report in excel format	Click on the “Excel” option	Downloading the forecasting report for all the years up to 2030 in excel format	Same as expected	Pass
13	Downloading Forecasting Report in pdf format	Click on the “Pdf” option	Downloading the forecasting report for all the years up to 2030 in pdf format	Same as expected	Pass
Testing Tourism Forecasting with Neural Network AutoRegression					
Testing Tab - Forecasting Plot					
14	Getting Predictions Using the NNAR model	Bring the sidebar near 30	Generating the NNAR plot for up to June 2022	Same as expected	Pass
Testing Tab - Forecasting Report by Year					
15	View Forecasting Reports by Year	Choose the 2025 as the year	Generating tourist arrivals forecast report for the year 2025	Same as expected	Pass
16	View the tourist arrivals for the rest	Click on the expand icon	Expanding the report and displaying the rest of the	Same as expected	Pass

	of the months which are not displayed in the page		months with the number of tourist arrivals		
17	Downloading Forecasting Report in excel format	Click on the “Excel” option	Downloading the forecasting report for the selected year in excel format	Same as expected	Pass
18	Downloading Forecasting Report in pdf format	Click on the “Pdf” option	Downloading the forecasting report for the selected year in pdf format	Same as expected	Pass
Testing Tab - Forecasting Report					
19	Search details using search bar	Search the year 2025	Displaying the tourist arrivals for each month of the year 2025	Same as expected	Pass
20	View the tourist arrivals for the rest of the months which are not displayed in the page	Click on the expand icon	Expanding the report and displaying the rest of the months with the number of tourist arrivals	Same as expected	Pass
21	Downloading Forecasting Report in excel format	Click on the “Excel” option	Downloading the forecasting report for all the years up to 2030 in excel format	Same as expected	Pass
22	Downloading Forecasting Report in pdf format	Click on the “Pdf” option	Downloading the forecasting report for all the years up to 2030 in pdf format	Same as expected	Pass
Testing Past Tourist Arrivals Reports					
Testing Tab - Tourist Arrivals by Year and Month					
23	View Past Reports based on the selected year	Choose the 2012 as the year	Generating a past tourist arrivals report for the year 2012	Same as expected	Pass

24	Search details using the search bar	Search the month December	Displaying the past tourist arrivals for December of the selected year	Same as expected	Pass
25	Downloading past report in excel format	Click on the “Excel” option	Downloading the past report for the selected year in excel format	Same as expected	Pass
26	Downloading past report in pdf format	Click on the “Pdf” option	Downloading the past report for the selected year in pdf format	Same as expected	Pass
Testing Tab - Tourist Arrivals by Continent, Region and Country					
27	View Past Reports based on the selected year, continent, region and country	Year: 2020 Continent: Europe Region: Western Europe Country: France	Generating past tourist arrivals report from France in 2020	Same as expected	Pass
28	Search details using the search bar	Search the month of December	Displaying the past tourist arrivals for December based on the selected inputs	Same as expected	Pass
29	Downloading past report in excel format	Click on the “Excel” option	Downloading the past report based on the selected inputs in excel format	Same as expected	Pass
30	Downloading past report in pdf format	Click on the “Pdf” option	Downloading the past report based on the selected inputs in pdf format	Same as expected	Pass

Table 7.1.1:Black-box testing

$$\text{Functional test pass rate } = \frac{\text{Number of tets passed}}{\text{Number of tests performed}} = \frac{30}{30} = 100\%$$

7.2 User Testing

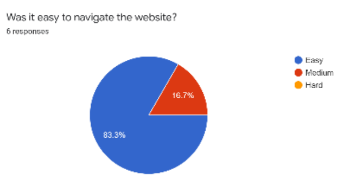
A questionnaire (refer to **Appendix F - Questionnaire for User Feedback**) was prepared to collect feedback from expert and non-expert users. The evaluation process has been divided into two phases where, after collecting feedback from the first evaluators, some changes have been made to the developed system based on their feedback. Then the feedback form was sent to the second set of evaluators to collect their feedback. The following **Table 7.2.1** will present the details of the selected experts and non-experts. Their feedback for the web-based tourism demand forecasting system will be displayed in **Table 7.2.2**

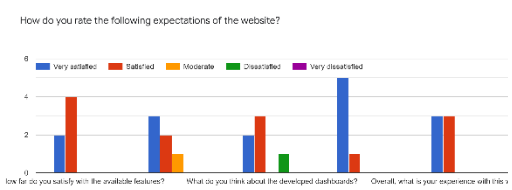
Note: [1] – first evaluators, [2] – second evaluators

Group	Name	Designation	Workplace
Experts			
Industrial Expert	Ravidu Perera [1]	Software Engineer	IFS R&D International Pvt Ltd
	Amila Haputhanthree [2]	Software Engineer	Creative Software
	Asini Pathmila Silva [2]	Software Engineer Intern	WSO2
Domain Expert	Aruna Prasanjeewa [2]	Junior Manager	SLTDA
	Sisira Kumara [2]	Senior Cashier	Hotel Sapphire
Non-Expert			
Sachin Wickramasuriya [1]		Undergraduate of RMIT University	

Table 7.2.1: Details of the selected experts and non-experts

Person	Feedback
Question: What do you think about the theme of the website?	
Ravidu Perera [1]	*“its professional. but I would go for some colurful, but calm theme”*
Sachin Wickramasuriya [1]	*“The visual theme is pleasing and seems fitting for a website of this purpose”*
Amila Haputhanthree [2]	*“Good”*
Asini Pathmila Silva [2]	*“Your website theme looks professional and it is user-friendly. The color theme that you used is good because it is so simple and easy to learn. The size and type of the font are also good”*

<table>
<tr><td>Aruna Prasanjeewa [2]</td><td>“Sri Lanka Tourism Forecasting System”</td></tr>
<tr><td>Sisira Kumara [2]</td><td>“I think it's really nice”</td></tr>
<tr><td colspan="2">Question: What do you think about the designed web interfaces? Was it easy to use?</td></tr>
<tr><td>Ravidu Perera [1]</td><td>“Easy to use and easy to understand. Better if you can add dashboard first”</td></tr>
<tr><td>Sachin Wickramasuriya [1]</td><td>“The web interface was easy enough to use and understand. However, including a 'How to use this website' tab in the top menu with step-by-step instructions could also be helpful to users”</td></tr>
<tr><td>Amila Haputhanthree [2]</td><td>“Yes”</td></tr>
<tr><td>Asini Pathmila Silva [2]</td><td>“Your website design follows Jakob Nielsen's usability heuristics. Your website design follows the visibility of the system status, consistency, standards, aesthetics, and minimalistic design. According to my experience, I think this is a good website design since you have followed the usability heuristics”</td></tr>
<tr><td>Aruna Prasanjeewa [2]</td><td>“Great Work, Please try to develop as a dash board and try to introduced the annual summery”</td></tr>
<tr><td>Sisira Kumara [2]</td><td>“Yes, it's easy to use”</td></tr>
<tr><td>Question: Was it easy to navigate the website?</td><td>Question: How do you rate the following expectations of the website?</td></tr>
<tr><td>


Five evaluators stated that the website was easy enough to navigate whereas one evaluator states it was medium.</td><td>


Two of the six evaluators are extremely satisfied with the available features, while the rest are satisfied. And three of them are really satisfied with how easy it is for them to get predictions using this system, while among the others, two are satisfied and one is moderate. One of the first evaluators is dissatisfied with them, whereas two are very satisfied and three are satisfied with the developed dashboard. Five are very satisfied with the offerings given by this system, and one is satisfied. Overall, three people are very satisfied with the system, while three others are satisfied.</td></tr>
</table>

Question: What do you think about this web application?	
Ravidu Perera [1]	*"It's better if you can start the app in dashboard. because lot of people don't like to sign up. so better to add guest sign at least"*
Sachin Wickramasuriya [1]	*"This web application has the potential to be very helpful to local and international businesses catering to tourists travelling to Sri Lanka"*
Amila Haputhanthree [2]	*"Informative and useful"*
Asini Pathmila Silva [2]	*"Overall, this website provides a broad view of Sri Lankan tourism demand. According to these features, I think this will be the only platform to get an idea of those predictions. So I think this website will be a great invention for Sri Lankan tourism"*
Aruna Prasanjeewa [2]	*"Need to Develop model the using the scenario as per situations like Easter attack, Covid pandemic situations and update the forecast every three years"*
Sisira Kumara [2]	*"This web application is very valuable to tourism related persons. According to this web application, tourism related persons can get ideas about how can they improve facilities which provided for tourists. We can provide great service for tourists, due to that we can attract more tourists to our country"*
Question: Do you have any suggestions for future enhancements?	
Ravidu Perera [1]	*"Improve the accuracy"*
Sachin Wickramasuriya [1]	*"A more suitable graph type for the 'Tourist Arrivals in Sri Lanka by Region' rather than the line graph. SLTDFS Logo on the top left-hand corner of the page. An explanation of how the two prediction models make the predictions and what makes them different. The dashboard for 'Past Tourist Arrivals' was a bit hard to understand at first-glance. Visually separating the sections that change depending on the selected geographical, from the other sections area could be helpful. In Past Tourist Arrivals Reports take out the 'Tourist Arrivals by Region and Country', 'Tourist Arrivals by Country' sections and just leave 'Tourist Arrivals by Continent, Region and Country' section; and add a 'Choose a country' option to that section."*
Asini Pathmila Silva [2]	*"It would be nice if you could use a lighter color scheme for the charts in the "Past Tourist Arrivals" and "Predicted Tourist Arrivals" dashboards"*

Aruna Prasanjeewa [2]	*"Great work and consider the above facts in future development"*

Table 7.2.2: Feedback of the expert and non-expert users

Some modifications to the Sri Lanka Tourism Demand Forecasting System were made after receiving feedback from the first evaluators before collecting feedback from the second evaluators. The following are the areas that have been modified as a result of the first evaluators' feedback:

- The website's white color theme has been replaced with a green color theme.
- The dashboards for past and predicted tourist arrivals have been redesigned.
- The line graph that has been used for the "Tourist Arrivals in Sri Lanka by Region" section on the home page was changed into a bar chart graph.
- The previously available past reports of the "Tourist Arrivals by Region and Country" and "Tourist Arrivals by Country" reports were removed and a "Choose a country" option was added to the "Tourist Arrivals by Continent, Region and Country" report.
- A new section, "How to Use" has been added to the navigation in order to explain how to get the predictions using the ARIMA and NNAR models.

Please refer to **Appendix G – Evidence of User Feedback**

8. CONCLUSIONS AND REFLECTIONS

8.1 Results of the application

Accuracy of the ARIMA model as follows,

```
Series: arrivals_timeseries
ARIMA(3,1,1)(0,1,2)[12]

Coefficients:
         ar1     ar2      ar3      ma1     sma1     sma2
      0.6715  0.1224  -0.1336  -0.9136  -0.1212  -0.1001
s.e.  0.0576  0.0553   0.0484   0.0388   0.0558   0.0567

sigma^2 = 67067050:  log likelihood = -5243.63
AIC=10501.25   AICc=10501.48   BIC=10530.79

Training set error measures:
                    ME     RMSE      MAE       MPE     MAPE      MASE       ACF1
Training set -148.0059 8037.259 4837.095 -1.534496 11.39433 0.5136693 0.00107158
```

Accuracy of the NNAR model as follows,

```
Forecast method: NNAR(14,1,8)[12]

Model Information:

Average of 20 networks, each of which is
a 14-8-1 network with 129 weights
options were - linear output units

Error measures:
                    ME     RMSE     MAE       MPE     MAPE      MASE       ACF1
Training set -10.17167 3833.311 2771.26 -2.026369 8.841423 0.2942906 0.00501099
```

Based on the accuracy of the ARIMA and NNAR models, **Table 8.1.1** below provides a summary of the results.

Model	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
ARIMA(3,1,1)(0,1,2)[12]	-148.0059	8037.259	4837.095	-1.534496	11.39433	0.5136693	0.00107158
NNAR(14,1,8)[12]	-10.17167	3833.311	2771.26	-2.026369	8.841423	0.2942906	0.00501099

Table 8.1.1: Accuracy of the trained models

According to **Table 8.1.1**, we can conclude that the **NNAR(14,1,8)[12]** model is the best model for forecasting tourist arrivals after the covid 19 outbreak because it has comparably lower evaluation measurement values than the ARIMA model. As a result, the NNAR(14,1,8)[12] model performs better than the ARIMA(3,1,1)(0,1,2)[12] model according to the trained models.

8.2 Strength and weakness of the application

The strength and weakness of the application have been discussed in the following **Table 8.2.1**.

Strengths	
A unique solution for the Sri Lankan context	The Sri Lanka Tourism Demand Forecasting System is a unique application in the Sri Lankan context that has not been implemented in the past decade and therefore would have significant value for the Sri Lankan tourism industry to manage their operations smoothly using this solution.
Easily accessible	Since this web-based tourism demand forecasting system is deployed on the internet, the Sri Lanka Tourism Development Authority and hotel managers will be able to access the web application anytime, anywhere.
Ability to get the tourist arrivals using different statistical forecasting models	Since the Sri Lanka Tourism Demand Forecasting System allows users to perform the tourism forecasting using ARIMA model and NNAR model, they can compare these two models and get the predicted tourist arrivals from each model.
Availability of informatic dashboards	Since two dashboards have been integrated with this Sri Lanka Tourism Demand Forecasting system, it will help Sri Lanka Development Authority

	and hotel managers make decisions quickly in order to improve their operational efficiency and productivity.
Availability of downloading reports	On the same platform, the Sri Lanka Tourism Development Authority and hotel managers will be able to download the past tourist arrivals details as well as the predicted tourist arrivals details. This will help to manage their operations smoothly.
Weaknesses	
Storing registration details to MySQL database	Since this web application was implemented using the R Shiny framework, it was really challenging to connect it with the MySQL database. Even though the web application could store the registration details in the database when a user registers, the shiny app should re-run in order to allow users to log in to the system using their registered credentials.
Accuracy of the predictions	Even though the research project aims to improve the accuracy of the tourist arrivals predictions, the trained models could not be able to achieve 100% accuracy for the predictions.

Table 8.2.1: Strengths and weaknesses of the application

8.3 Acquisition of new knowledge and skills

The new knowledge and skills that were acquired throughout the project have been discussed below **Table 8.3.1**.

Knowledge	
Time series forecasting	Before undertaking this research project, I had no prior knowledge of time series forecasting. However, since tourist arrivals forecasting belongs to the field of time series forecasting, it helped me gain a wealth of knowledge about this field as a result of this research.
Machine Learning	I have never learned about one of the traditional machine learning algorithms of ARIMA and artificial neural network type of NNAR algorithms before developing this project. But now I have knowledge regarding how to build these models in order to make the predictions using these models.
Skills	

R Shiny Web Development	Since the Sri Lanka Tourism Demand Forecasting System was developed using R Shiny, I have developed a new skill of developing web apps using R Shiny, which I have never used before.
Data Visualization with Power BI	Because of designing dashboards for this research project, I could be able to develop a new skill of data visualization using Power BI that I have never used before.

Table 8.3.1: Acquisition of new knowledge and skills

8.4 Further Work

Despite the fact that this Sri Lanka Tourism Demand Forecasting System was successfully developed within the time frame given, there are still some aspects that could be further improved. Due to the time constraints, these additional tasks were not developed, and the project scope was determined to be sufficient to complete the project on time. On the other hand, as a Business Information Systems student, I have little technical knowledge regarding machine learning and other technological aspects. Therefore, it was decided to choose the scope that is most appropriate for my knowledge and capability. The further work that can be addressed is as follows,

- The accuracy of the predictions can be further improved using different forecasting models since this research project only looked at statistical forecasting models like ARIMA and NNAR.
- Predictions can be further calculated by developing models considering scenarios such as civil war and easter attack, in addition to covid pandemic situations.
- The predictions that were manually calculated by the Sri Lanka Tourism Development Authority could be integrated with this Sri Lanka Tourism Demand Forecasting System as a feature in order to allow them to adjust predictions according to their manual trend analysis.

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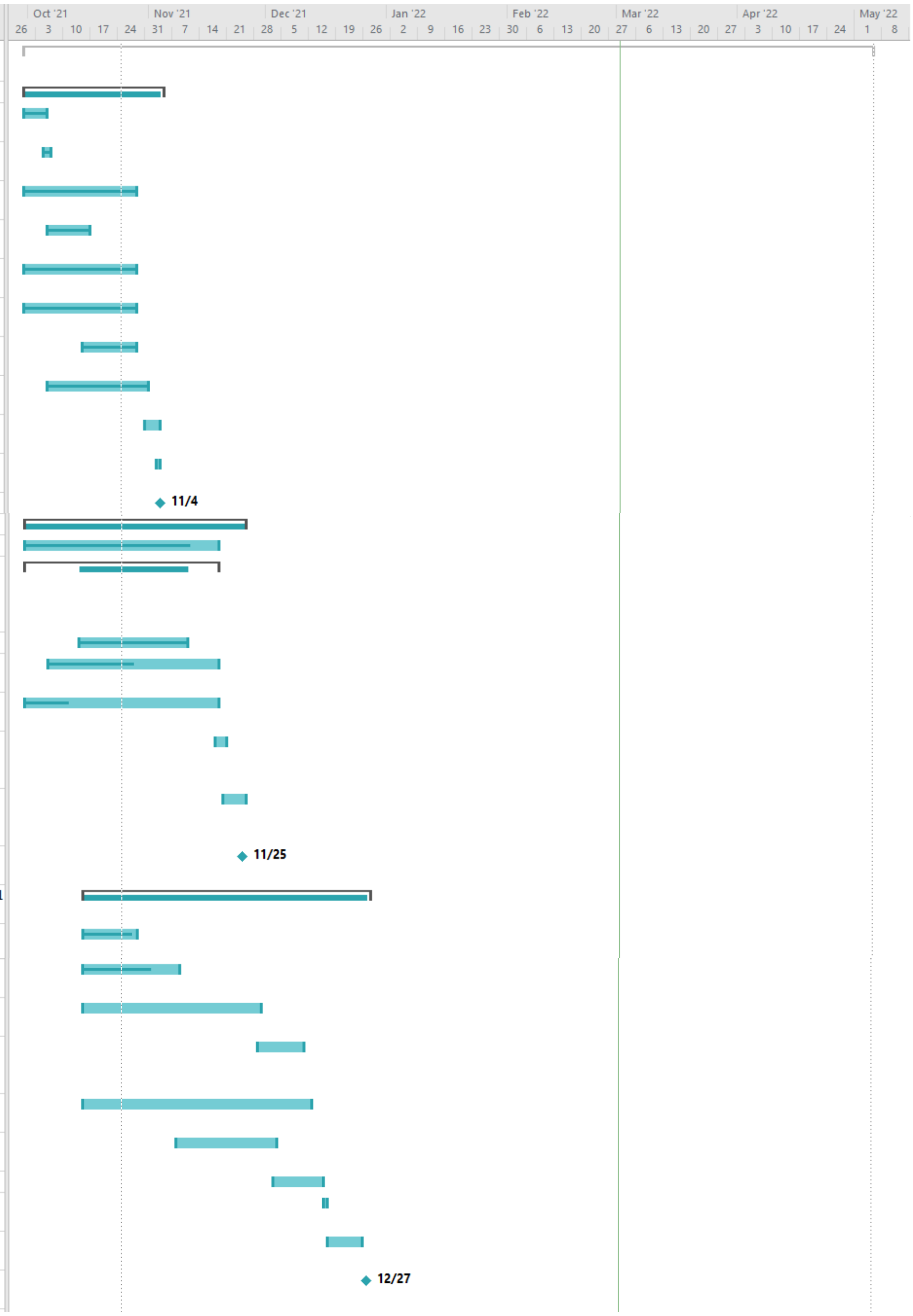
Appendix A – Detailed Gantt Chart

GANTT CHART

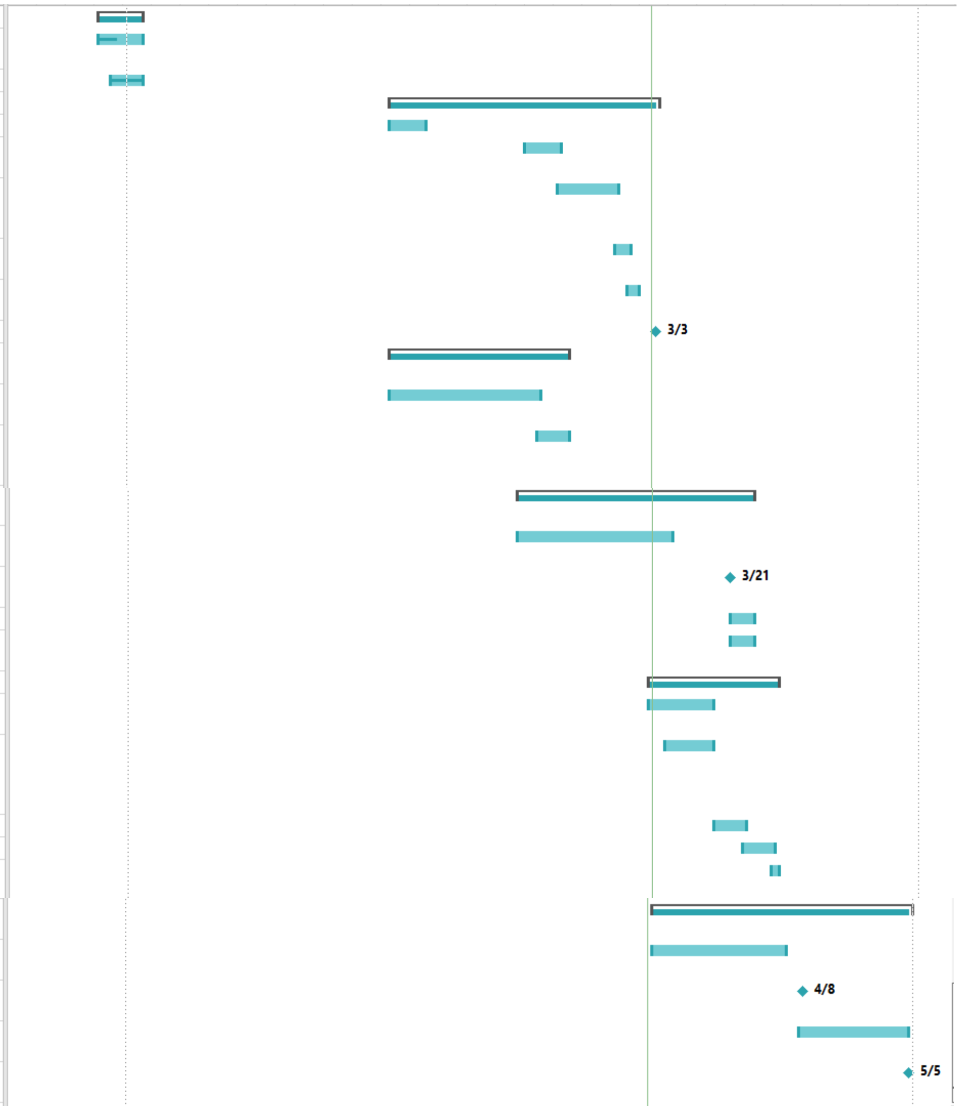
	Task Name	Start	Finish
0	**Final Year Project Timeline**	**Thu 9/30/21**	**Thu 5/5/22**
1	**1 Project Initiation**	**Thu 9/30/21**	**Thu 11/4/21**
2	1.1 Selection of research problem and	Thu 9/30/21	Tue 10/5/21
3	1.2 Mentor approval of the research topic	Tue 10/5/21	Wed 10/6/21
4	1.3 Familiarizing with research domain	Thu 9/30/21	Thu 10/28/21
5	1.4 Identifying aim and objectives	Wed 10/6/21	Sat 10/16/21
6	1.5 Performing initial literature review	Thu 9/30/21	Thu 10/28/21
7	1.6 Identify the necessary tools and	Thu 9/30/21	Thu 10/28/21
8	1.7 Documenting initial list of	Fri 10/15/21	Thu 10/28/21
9	1.8 Preparing draft Project Proposal (PP)	Wed 10/6/21	Sun 10/31/21
10	1.9 Mentor and Peer review of draft PP	Sun 10/31/21	Wed 11/3/21
11	1.10 Self-evaluation and refinement of PP	Wed 11/3/21	Wed 11/3/21
12	1.11 Submission of PP	Thu 11/4/21	Thu 11/4/21
13	**2 Literature Review**	**Thu 9/30/21**	**Thu 11/25/21**
14	2.1 Literature Survey	Thu 9/30/21	Thu 11/18/21
15	**2.2 Literature Review of the Research Projects/ Applications (Similar & Relevant)**	**Thu 9/30/21**	**Thu 11/18/21**
16	2.2.1 Competitor An	Thu 10/14/21	Wed 11/10/21
17	2.3 Literature Review of Tools and	Wed 10/6/21	Thu 11/18/21
18	2.4 Prepare draft Literature Review	Thu 9/30/21	Thu 11/18/21
19	2.5 Mentor and Peer review of draft Literature Review	Thu 11/18/21	Sat 11/20/21
20	2.6 Self-evaluating and refining Literature Review	Sat 11/20/21	Thu 11/25/21
21	2.7 Submission of final Literature	Thu 11/25/21	Thu 11/25/21
22	**3 Requirement Specification**	**Fri 10/15/21**	**Mon 12/27/21**
23	3.1 Identifying Stakeholders	Fri 10/15/21	Thu 10/28/21
24	3.2 Analyzing current processes	Fri 10/15/21	Mon 11/8/21
25	3.3 Gathering Requirements	Fri 10/15/21	Mon 11/29/21
26	3.4 Modeling Requirements using relevant diagrams	Mon 11/29/21	Fri 12/10/21
27	3.5 Project Requirements	Fri 10/15/21	Sun 12/12/21
28	3.6 Legal, social and ethical issues	Mon 11/8/21	Fri 12/3/21
29	3.7 Creating draft SRS	Fri 12/3/21	Wed 12/15/21
30	3.8 Mentor and peer review of SRS	Thu 12/16/21	Thu 12/16/21
31	3.9 Self-Evaluation and refinement of SRS	Fri 12/17/21	Sat 12/25/21
32	3.10 Submission of Final SRS	Mon 12/27/21	Mon 12/27/21



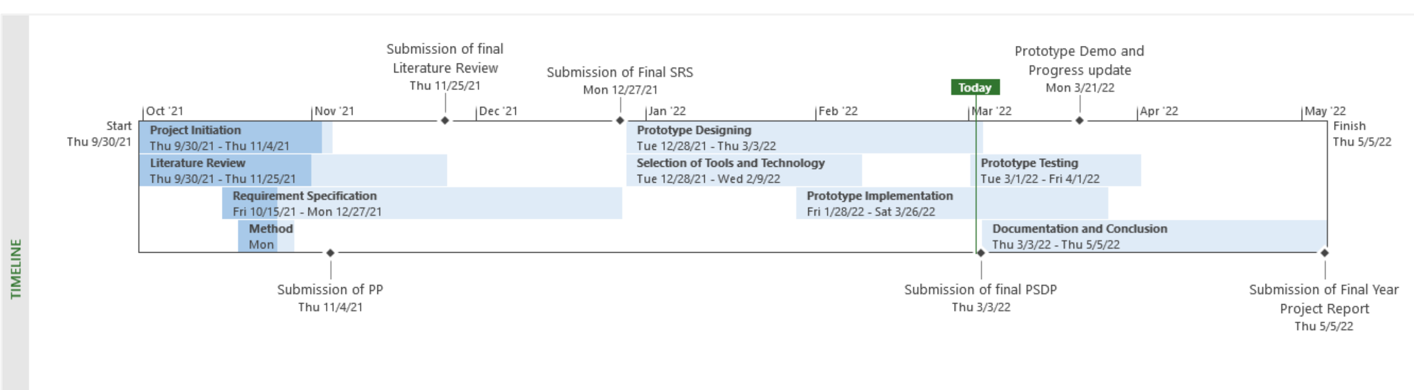
GANTT CHART

	Task Name	Start	Finish
33	**4 Methodologies**	**Mon 10/18/21**	**Thu 10/28/21**
34	4.1 Selecting Methodologies	Mon 10/18/21	Thu 10/28/21
35	4.2 Work Plan	Thu 10/21/21	Thu 10/28/21
36	**5 Prototype Designing**	**Tue 12/28/21**	**Thu 3/3/22**
37	5.1 Identifying compor	Tue 12/28/21	Wed 1/5/22
38	5.2 Creating the prototype design	Sun 1/30/22	Mon 2/7/22
39	5.3 Preparing draft of Project Specifications Design and Prototype	Mon 2/7/22	Mon 2/21/22
40	5.4 Mentor and Peer review of draft PSDP	Mon 2/21/22	Thu 2/24/22
41	5.5 Self-evaluating and refining PSDP	Thu 2/24/22	Sat 2/26/22
42	5.6 Submission of final	Thu 3/3/22	Thu 3/3/22
43	**6 Selection of Tools and Technology**	**Tue 12/28/21**	**Wed 2/9/22**
44	6.1 Reviewing different tools and	Tue 12/28/21	Wed 2/2/22
45	6.2 Selection of appropriate tools and technology	Wed 2/2/22	Wed 2/9/22
46	**7 Prototype Implementation**	**Fri 1/28/22**	**Sat 3/26/22**
47	7.1 Implementing core functionalities of	Fri 1/28/22	Sun 3/6/22
48	7.2 Prototype Demo and Progress update	Mon 3/21/22	Mon 3/21/22
49	7.3 Refining prototype	Mon 3/21/22	Sat 3/26/22
50	7.4 List of refined requirements	Mon 3/21/22	Sat 3/26/22
51	**8 Prototype Testing**	**Tue 3/1/22**	**Fri 4/1/22**
52	8.1 Identifying components to test	Tue 3/1/22	Wed 3/16/22
53	8.2 Identifying functional and nonfunctional testing	Sat 3/5/22	Wed 3/16/22
54	8.3 Functional Testing	Thu 3/17/22	Thu 3/24/22
55	8.4 User Testing	Thu 3/24/22	Thu 3/31/22
56	8.5 Final evaluation of prototype	Thu 3/31/22	Fri 4/1/22
57	**9 Documentation and Conclusion**	**Thu 3/3/22**	**Thu 5/5/22**
58	9.1 Creating draft FYP Report	Thu 3/3/22	Mon 4/4/22
59	9.2 Submission of FYP draft Report	Fri 4/8/22	Fri 4/8/22
60	9.3 Refining Final Year Project Report	Fri 4/8/22	Wed 5/4/22
61	9.4 Submission of Final Year Project	Thu 5/5/22	Thu 5/5/22

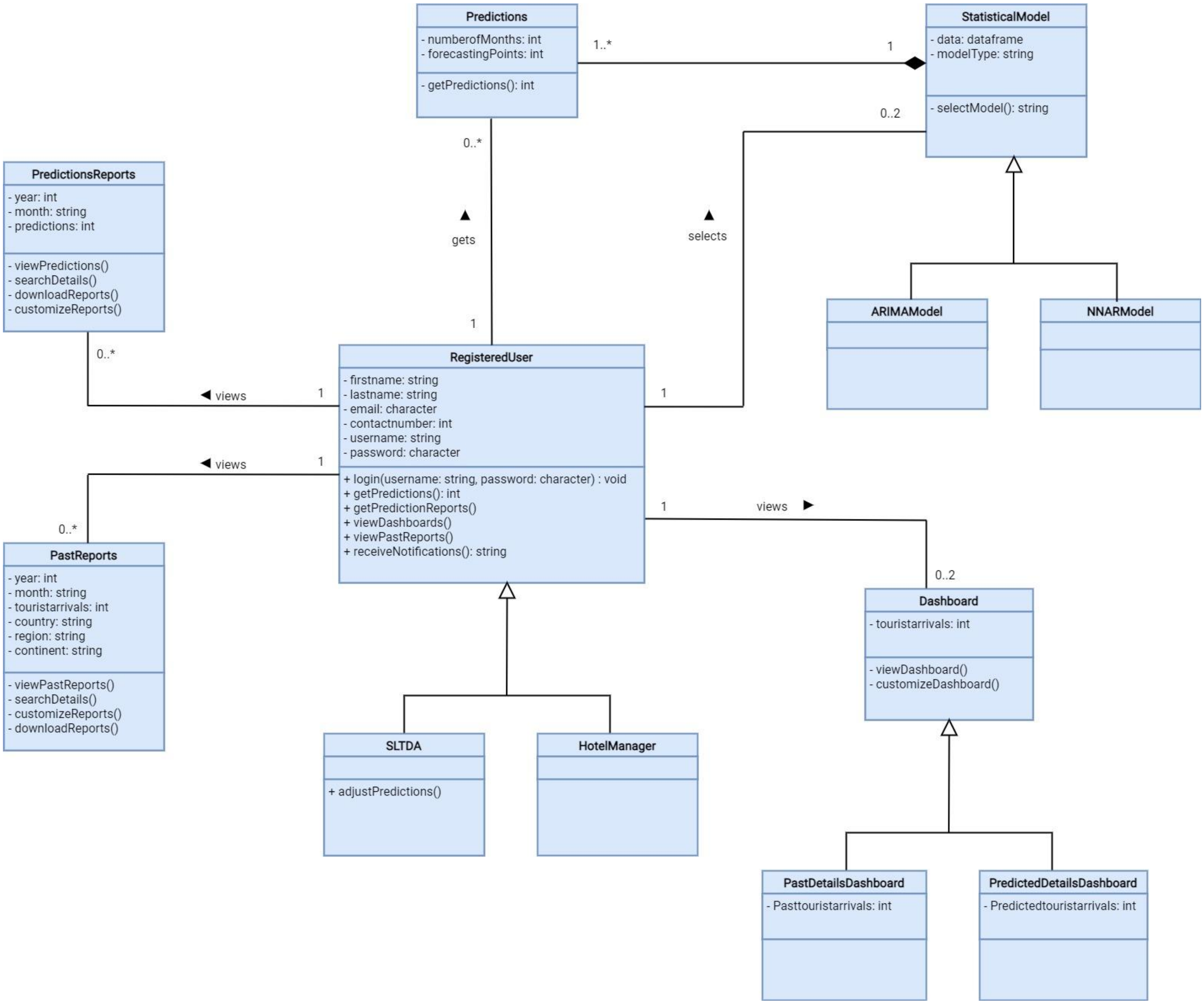
GANTT CHART



Appendix B - Timeline



Appendix C – Class Diagram



Appendix D – Other Functionalities

Displaying home page graphs

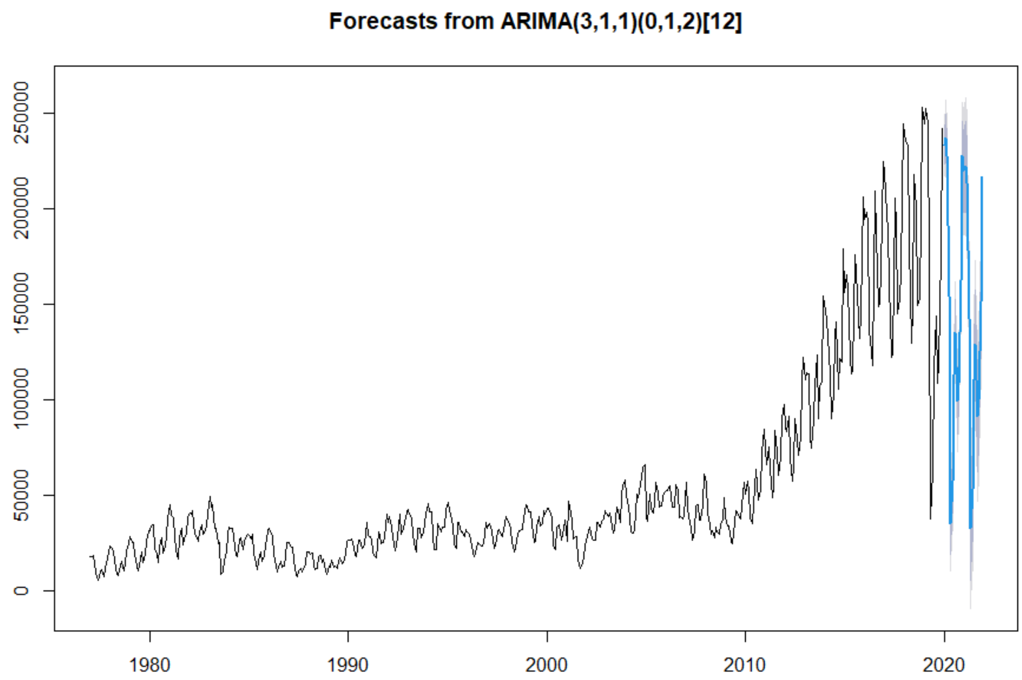
There are two types of graphs displayed on the home page in two tabs called, Tourist Arrivals in Sri Lanka and Tourist Arrivals in Sri Lanka by Region. These graphs were developed as follows,

```
#------------------------home page plots-----------------------------------------

month <- c('January', 'February', 'March', 'April', 'May', 'June', 'July',
           'August', 'September', 'October', 'November', 'December')
y_2020 <- arrivals$Tourist_Arrivals[517:528]
y_2021 <- arrivals$Tourist_Arrivals[529:540]
y_2022 <- y2022_arrivals$Tourist_Arrivals

data <- data.frame(month, y_2020, y_2021, y_2022)

#To alphabetized the order
data$month <- factor(data$month, levels = data[["month"]])

fig <- plot_ly(data, x = ~month, y = ~y_2020, name = '2020', type = 'scatter', mode = 'lines',
               line = list(color = 'rgb(205, 12, 24)', width = 4))
fig <- fig %>% add_trace(y = ~y_2021, name = '2021', line = list(color = 'rgb(22, 96, 167)', width = 4))
fig <- fig %>% add_trace(y = ~y_2022, name = '2022', line = list(color = 'rgb(0,128,0)', width = 4))
fig <- fig %>% layout(xaxis = list(title = "Months"),
                      yaxis = list (title = "Number of Tourist Arrivals"))


continent <- c('Americas', 'Africa', 'Asia & Pacific', 'Europe', 'Middle East')
yr_2020 <- cont_arrivals$Tourist_Arrivals[1:5]
yr_2021 <- cont_arrivals$Tourist_Arrivals[6:10]

data_2 <- data.frame(continent, yr_2020, yr_2021)

#The default order will be alphabetized unless specified as below:
data_2$continent <- factor(data_2$continent, levels = data_2[["continent"]])

fig_2 <- plot_ly(data_2, x = ~continent, y = ~yr_2020, name = '2020', type = 'bar')
fig_2 <- fig_2 %>% add_trace(y = ~yr_2021, name = '2021')
fig_2 <- fig_2 %>% layout(xaxis = list(title = "Region"),
                          yaxis = list (title = "Number of Tourist Arrivals"))
```

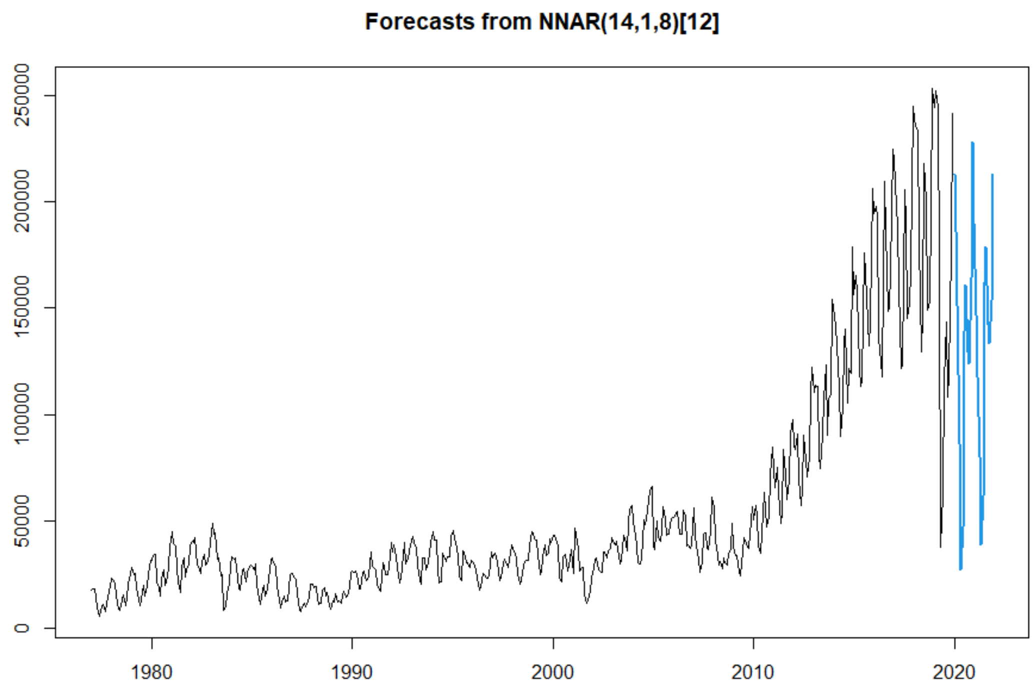
After developing the graphs, they have been included on the home page as follows,

```
tabsetPanel(
  tabPanel("Tourist Arrivals in Sri Lanka", br(), fig),
  tabPanel("Tourist Arrivals in Sri Lanka by Region", br(), fig_2),
),
```

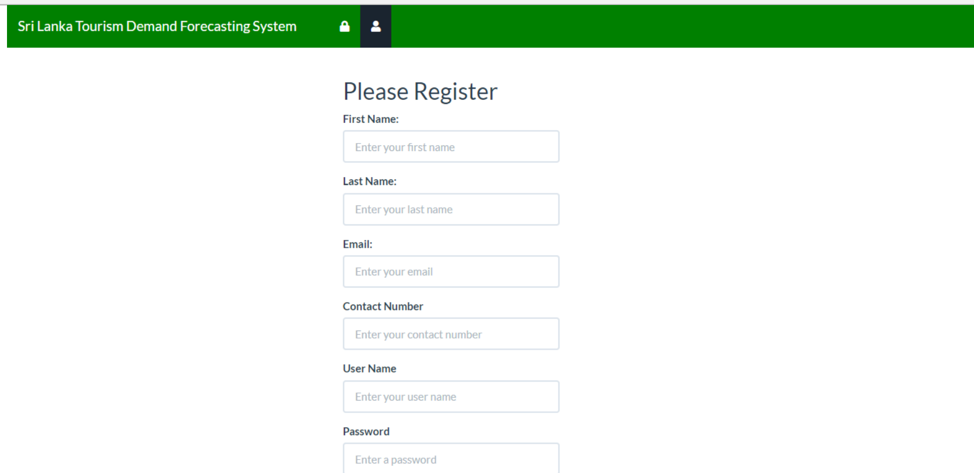
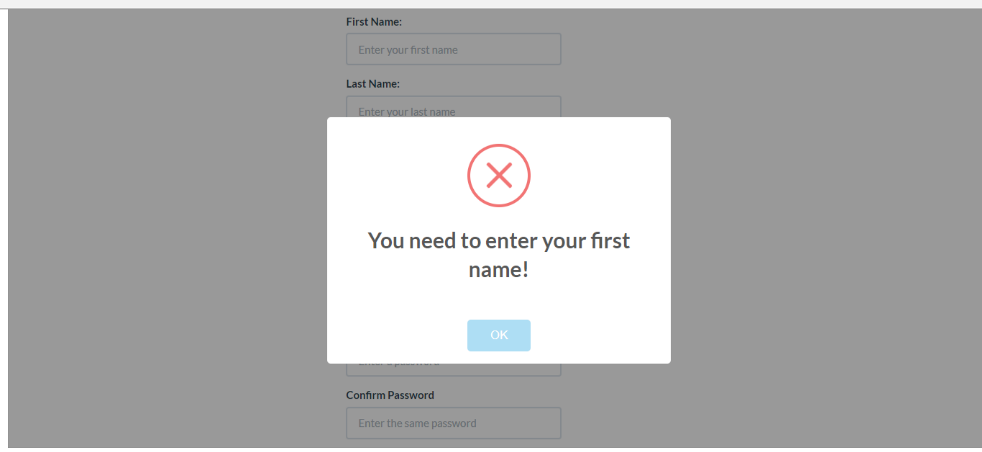
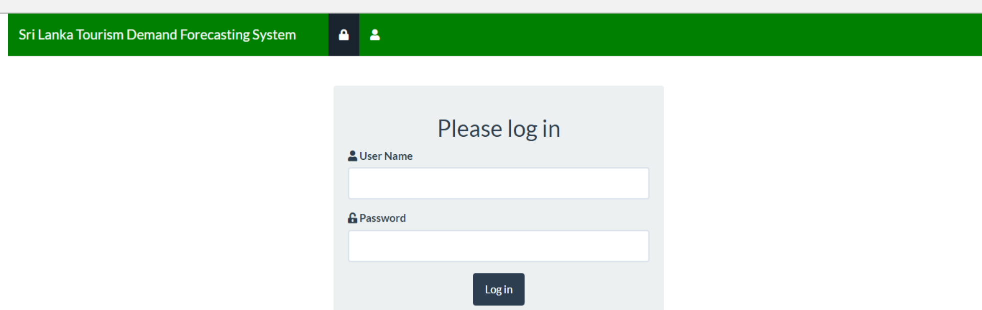
ARIMA Plot

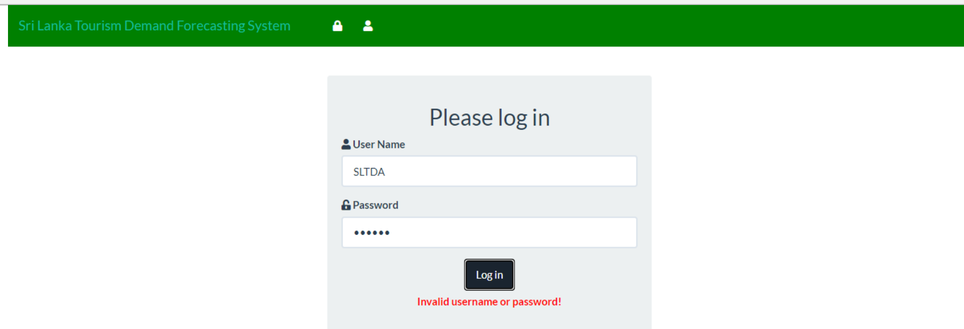
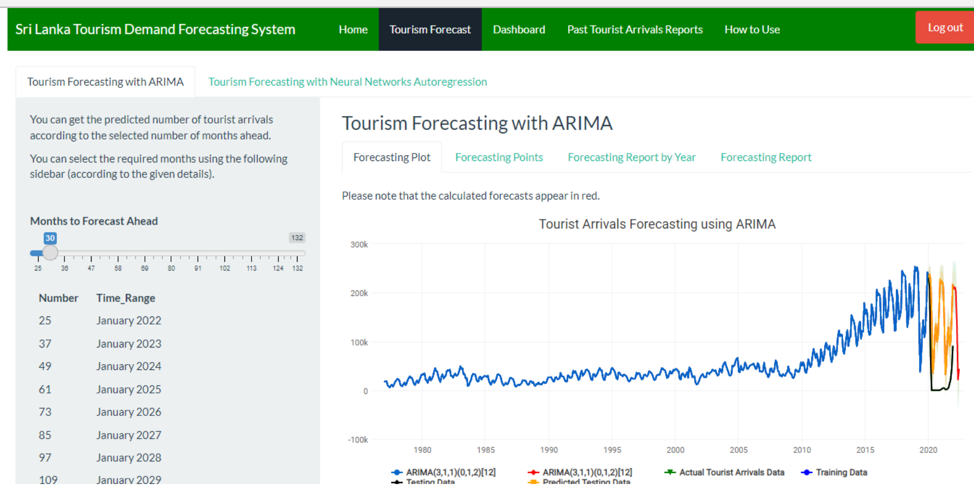
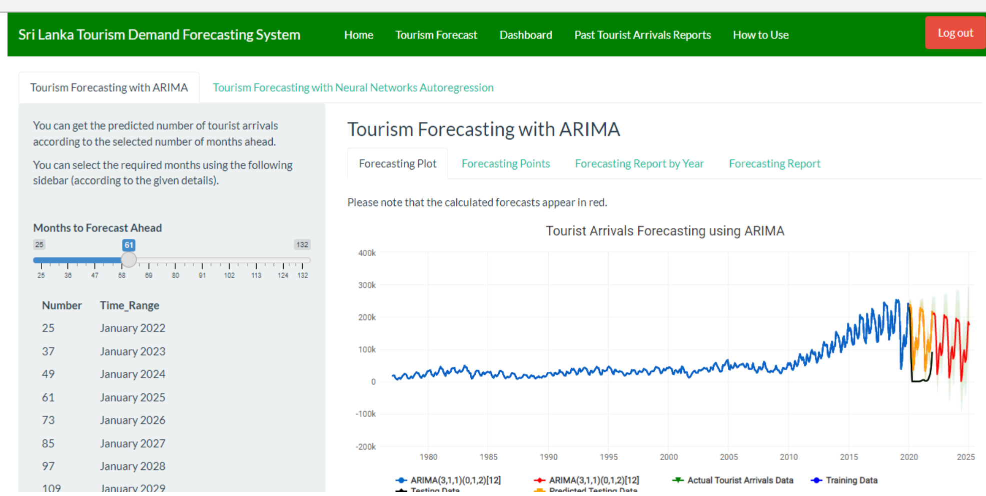


NNAR Plot

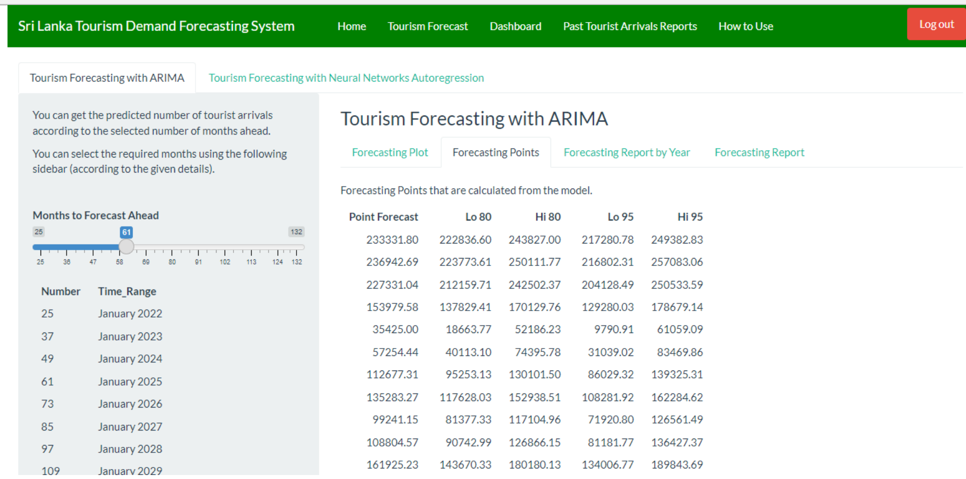


Appendix E – Requirement ID and its developed user interface

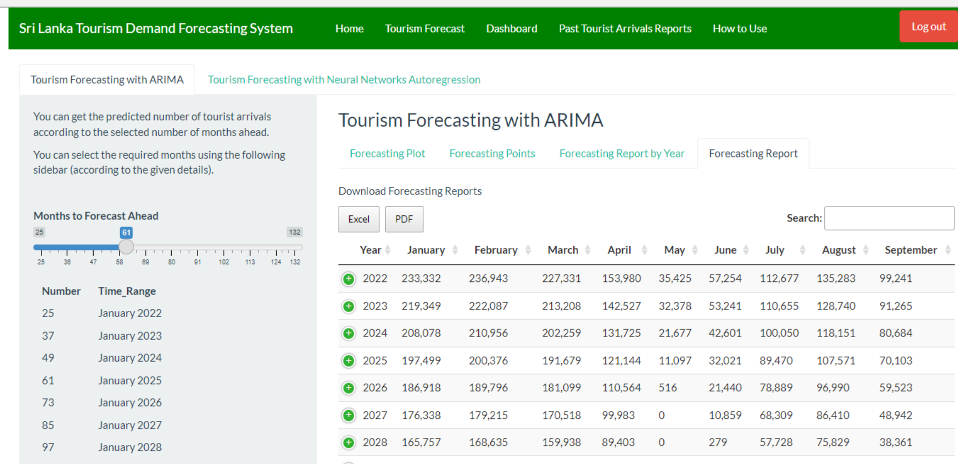
Req ID	Developed User Interface
FR1	**Register** 
FR2 FR3	**Validate user inputs/ Display Input field errors** 
FR4	**Login** 

FR5	**Verify user credentials/ Display Login error**
FR6	
FR7	**Select different statistical forecasting models**
	
FR8	**Select time range**
	

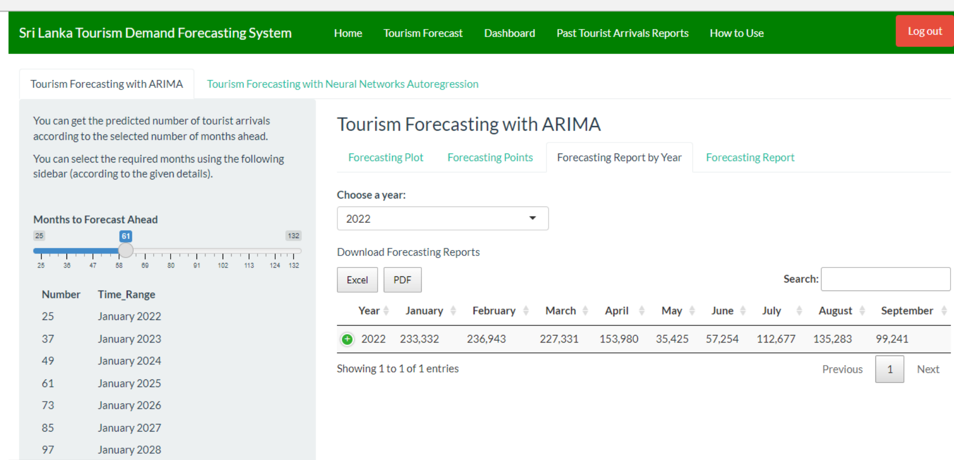
FR9	**Get predicted tourist arrivals**



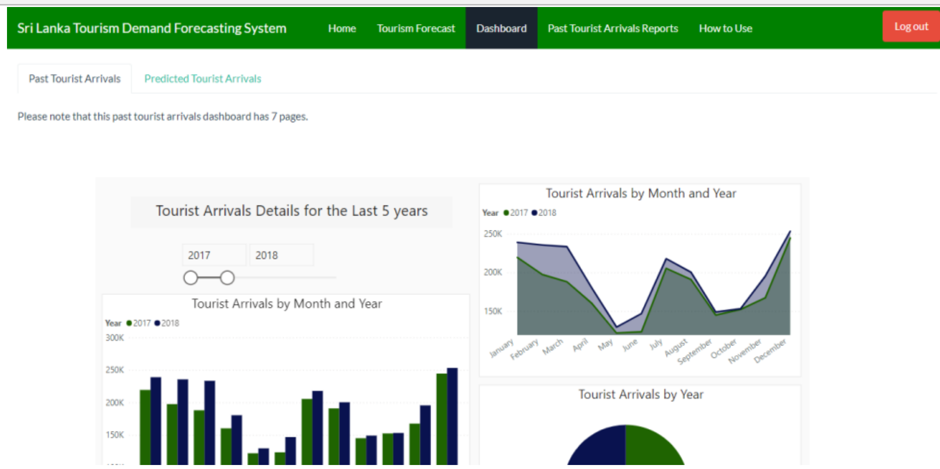
FR10 FR11	**View/ Download predicted tourist arrivals reports**

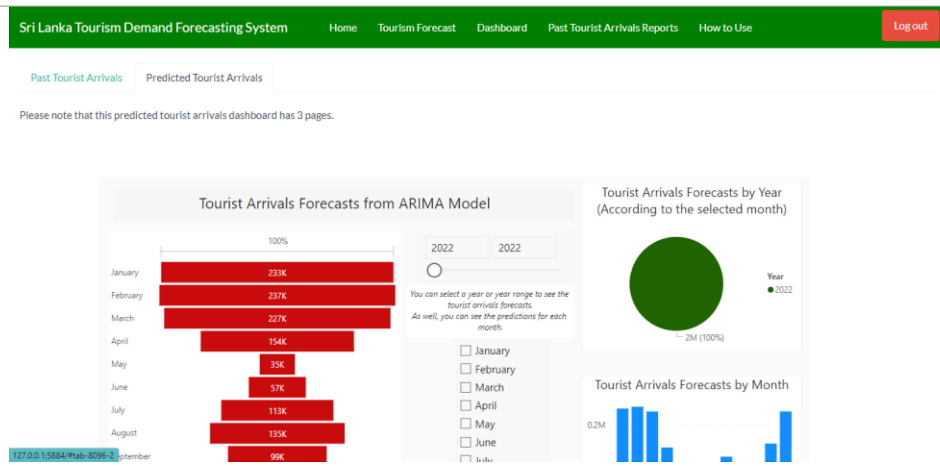


FR12 FR13	**Customize predicted tourist arrivals reports/ Download customized predicted tourist arrivals reports**

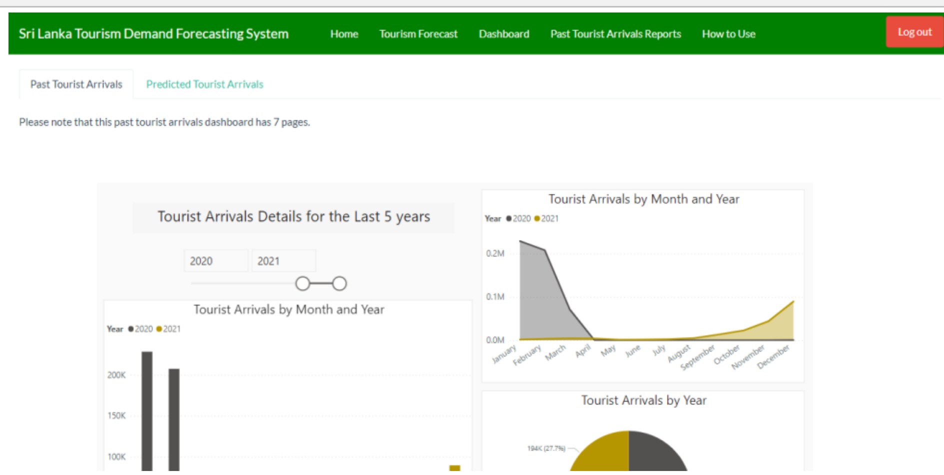


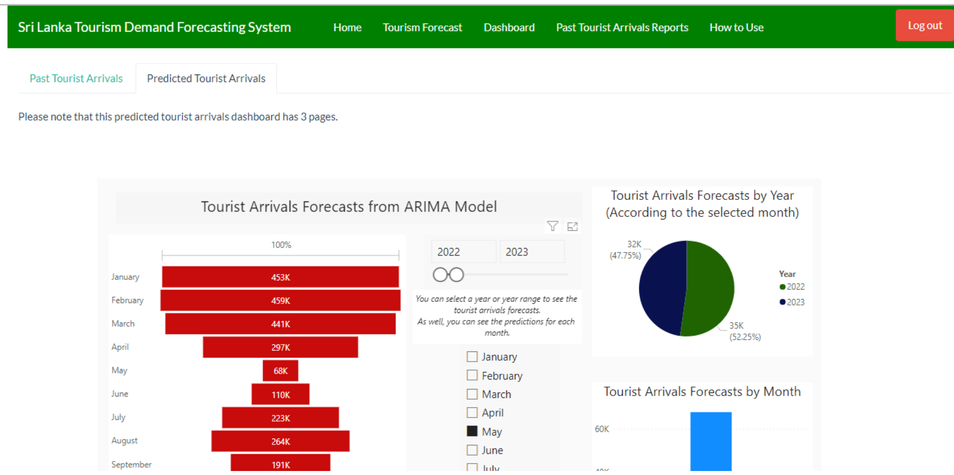
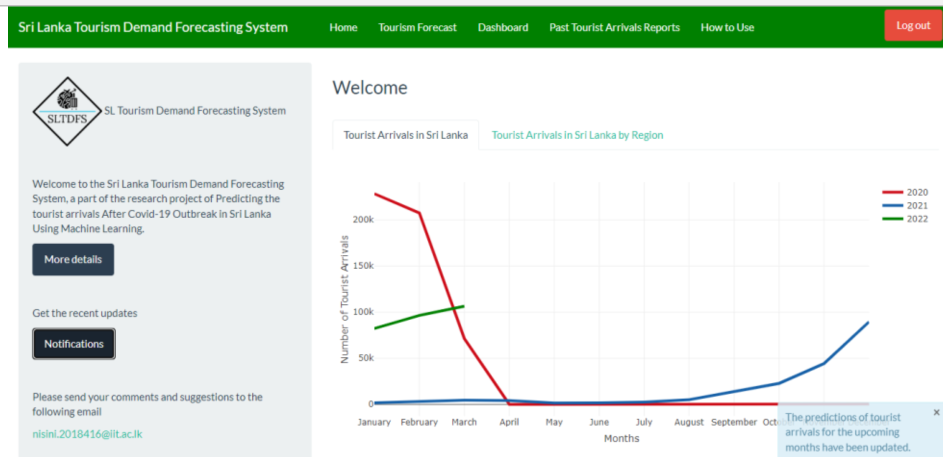
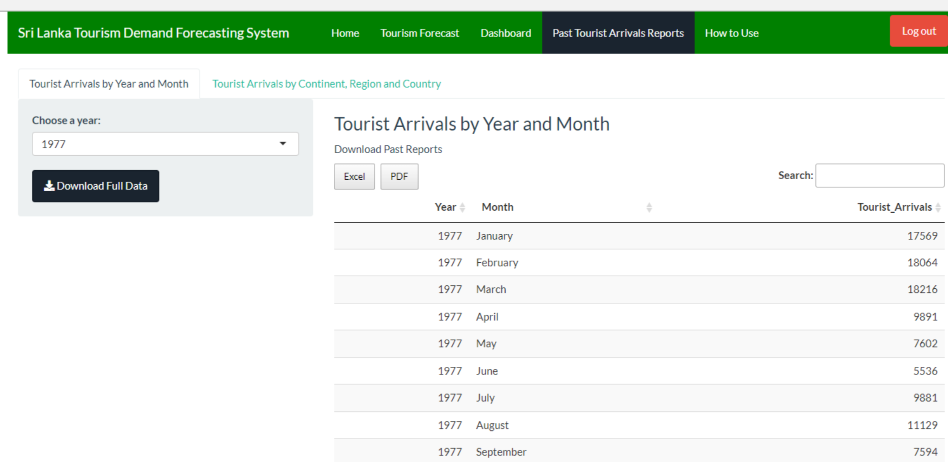
FR15 **View dashboards**





FR16 **Customize dashboards**



<table>
<tr><td></td><td>
</td></tr>
<tr><td>FR17</td><td>Make suggestions/ Receive notifications</td></tr>
<tr><td>FR18</td><td>
</td></tr>
<tr><td>FR19</td><td>View/ Download customized past tourist arrivals reports</td></tr>
<tr><td>FR20</td><td>
</td></tr>
</table>

Appendix F - Questionnaire for User Feedback

Section 1 of 3

Sri Lanka Tourism Demand Forecasting System Feedback

Form description

Email *

Valid email

This form is collecting emails. Change settings

Please select your language. කරුණාකර ඔබේ භාෂාව තෝරන්න. *

- English
- සිංහල (Sinhala)

Section 2 of 3

Sri Lanka Tourism Demand Forecasting System Feedback

Hello! I'm Nisini Silva. I am currently working on a research study on predicting tourist arrivals after the covid-19 outbreak in Sri Lanka, which provides a web-based tourism demand forecasting system that can predict the number of tourist arrivals for the upcoming months. The aim of this survey is to collect feedback from you in terms of what you think about this solution and how it can be improved from your perspective.
Please be kind enough to spare a few minutes to fill out this form. Thank you in advance!

You can access the developed web-based tourism demand forecasting system using the below URL link.
https://sltdfs.shinyapps.io/Sri_Lanka_Tourism_Demand_Forecasting_System/

Please make sure to log in to the system using the following user credentials.
User Name - user1
Password - pass1

Please login to the web application using your computer or any other desktop application to get a better user experience.

Please note that your feedback will be included into my research project report.

Sri Lanka Tourism Demand Forecasting System Application Demo and Explanation

Your Name *

Your answer

Your designation *

Your answer

Workplace *

Your answer

What do you think about the theme of the website? *

Your answer

What do you think about the designed web interfaces? Was it easy to use? *

Your answer

Was it easy to navigate the website? *

- Easy
- Medium
- Hard

How do you rate the following expectations of the website? *

	Very satisfied	Satisfied	Moderate	Dissatisfied	Very dissatisfied
How far do you satisfy with the available features?	○	○	○	○	○
How easy was it to get the predictions using this web solution?	○	○	○	○	○
What do you think about the developed dashboards?	○	○	○	○	○
What do you think about the facilities (Ex: downloading reports) provided by this website	○	○	○	○	○
Overall, what is your experience with this web solution?	○	○	○	○	○

What do you think about this web application? *

Your answer

Do you have any suggestions for future enhancements?

Your answer

Appendix G – Evidence of User Feedback

Sri Lanka Tourism Demand Forecasting System Feedback

Your Name

6 responses

Aruna Prasanjeewa

Sachin Wickramasuriya

Sisira Kumara

Ravidu Perera

Amila

Asini Pathmila Silva

Your designation

6 responses

Junior Manager

Mr.

Senior Cashier

Software Engineer

Software engineer

Software Engineer Intern

Workplace

6 responses

Sri Lanka Tourism Development Authority

RMIT University

Hotel Sapphire

IFS R&D International Pvt Ltd

Creative Software

WSO2

What do you think about the theme of the website?

6 responses

Sri Lanka Tourism Forecasting System

The visual theme is pleasing and seems fitting for a website of this purpose.

I think it's really nice

its professional. but i would go for some colurful , but calm theme

Good

Your website theme looks professional and it is user-friendly. The color theme that you used is good because it is so simple and easy to learn. The size and type of the font are also good.

What do you think about the designed web interfaces? Was it easy to use?

6 responses

Great Work, Please try to develop as a dash board and try to introduced the annual summery

The web interface was easy enough to use and understand. However, including a 'How to use this website' tab in the top menu with step-by-step instructions could also be helpful to users.

Yes,it's easy to use

Easy to use and easy to understand. Better if you can add dashboard first

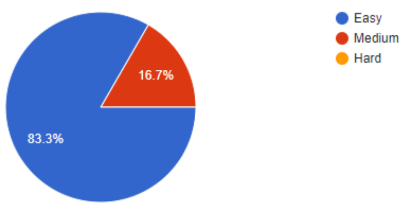
Yes.

Your website design follows Jakob Nielsen's usability heuristics. Your website design follows the visibility of the system status, consistency, standards, aesthetics, and minimalistic design. According to my experience, I think this is a good website design since you have followed the usability heuristics.

Was it easy to navigate the website?

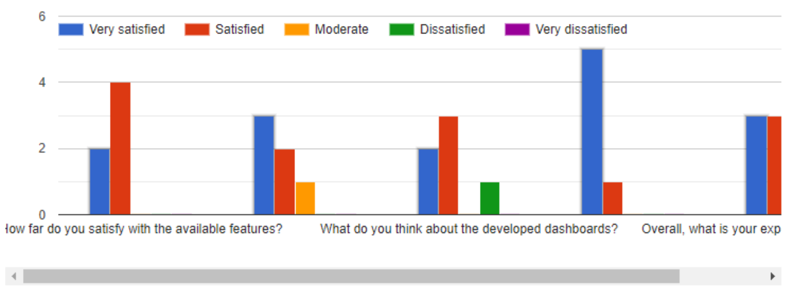
Copy

6 responses



How do you rate the following expectations of the website?

Copy



What do you think about this web application?

6 responses

Need to Develop model the using the scenario as per situations like Easter attack, Covid pandemic situations and update the forecast every three years

This web application has the potential to be very helpful to local and international businesses catering to tourists travelling to Sri Lanka.

This web application is very valuable to tourism related persons.
According to this web application,tourism related persons can get ideas about how can they improve facilities which provided for tourists.
We can provide great service for tourists,due to that we can attract more tourists to our country.

Its better if you can start the app in dashboard. because lot of people dont like to signup. so better to add guest sign atleast

Informative and useful

Overall, this website provides a broad view of Sri Lankan tourism demand. According to these features, I think this will be the only platform to get an idea of those predictions. So I think this website will be a great

Do you have any suggestions for future enhancements?

5 responses

Great work and consider the above facts in future development

A more suitable graph type for the 'Tourist Arrivals in Sri Lanka by Region' rather than the line graph.

SLTDFS Logo on the top left-hand corner of the page.

An explanation of how the two prediction models make the predictions and what makes them different.

The dashboard for 'Past Tourist Arrivals' was a bit hard to understand at first-glance. Visually separating the sections that change depending on the selected geographical, from the other sections area could be helpful.

In Past Tourist Arrivals Reports take out the 'Tourist Arrivals by Region and Country', 'Tourist Arrivals by Country' sections and just leave 'Tourist Arrivals by Continent, Region and Country' section; and add a 'Choose a country' option to that section.

I have no idea

improve the accuracy

Appendix H - On-line Video Demo of the project

https://youtu.be/G-wighJ9rQM

Appendix I – Project Logbook

Name of the Supervisee	E. Nisini Nethmila Silva
Student ID	2018416
Project Title	Predicting the Tourist Arrivals After Covid-19 Outbreak in Sri Lanka Using Machine Learning

Meeting #	Date	Time	Discussion/Guidance given	Tasks to complete before next meeting	Next Meeting Date	Next Meeting Time	Comments on previously assigned work	Comments on progress as per plan	Supervisor sign
1st Meeting	09/16/2021	From 8.00 p.m. to 9.00 p.m.	I was given guidance on how to change the project's topic and improve its usability by focusing on the project's business value.	Rephrasing the project title, determining the business value, and improving the project's usability, or considering another research domain.	09/23/2021	9.30 p.m.	NA	NA	S P Pitigala
2nd Meeting	09/23/2021	From 9.30 p.m. to 10.00 p.m.	I was guided by the lack of sufficient areas of the new project idea proposed. And I got some suggestions on how to develop that new project idea from several perspectives.	Finding the relevant data set from the previous ten years.	09/28/2021	10.30 p.m.	Completed satisfactory background study.	Satisfied.	S P Pitigala
3rd Meeting	09/28/2021	From 10.30 p.m. to 10.45 p.m.	Discussion of research opportunities based on the found dataset and identification of missing data that may be essential to the project's development. Also, I got assistance to find the most relevant data for my project.	Finding the relevant data set that was discussed in the meeting.	09/30/2021	9.00 p.m.	Difficulty of finding the relevant dataset was addressed in the meeting.	Satisfactory progress has been made.	S P Pitigala
4th Meeting	09/30/2021	From 9.00 p.m. to 9.20 p.m.	Other research domains that are sufficient for the research projects are discussed.	Consider the new ideas that have been suggested, and study previous research articles that have been written for them.	10/06/2021	9.30 p.m.	Satisfied.	Satisfied.	S P Pitigala
5th Meeting	10/06/2021	From 9.30 p.m. to 9.50 p.m.	The project concept idea was confirmed and the most important details related to the research study were discussed.	Prepare the project document, complete the introduction section and get feedback on it.	10/12/2021	9.30 p.m.	Finalised the project idea.	Started the writeup.	S P Pitigala
6th Meeting	10/12/2021	From 9.30 p.m. to 9.50 p.m.	Discussion about the feedback of the peer presentation.	Improve the fish-born diagram and find similar web applications.	10/14/2021	9.30 p.m.	Explore the possibilities of improving the project design.	Satisfied.	S P Pitigala
7th Meeting	10/14/2021	From 9.30 p.m. to 9.50 p.m.	Got feedback for improved fishborn diagram and discuss similar web applications.	Go through similar web applications. Add few more sub problems to the fish-born diagram. Complete the first section of the project proposal.	10/25/2021	10.00 p.m.	Strengthens and Weakness of similar kind system were discussed.	Satisfied.	S P Pitigala
8th Meeting	10/25/2021	From 10.00 p.m. to 10.30 p.m.	Discussion on how features can be improved in the proposed system.	Submission of completed sections of PID draft.	11/5/2021	9.00 p.m.	Satisfactory progress has been made.	Satisfied.	S P Pitigala
9th Meeting	11/5/2021	From 9.00 p.m. to 9.20 p.m.	Discussion on how to conduct preliminary analysis and literature review part	Prepare the interview questions and conduct the interviews. Start to draft on Literature Review	12/27/2021	6.30 p.m.	LR was started and need to cover more research papers.	Satisfied.	S P Pitigala
10th Meeting	12/27/2021	From 6.30 p.m. to 7.10 p.m.	Discussion on feedback for SRS	Update the SRS document according to the feedback	1/21/2022	9.00 p.m.	Review the SRS.	LR was satisfactory.	S P Pitigala

Note

(blue)	To be filled by the supervisee
(pink)	To be filled by the supervisor

11th Meeting	1/21/2022	From 9.00 p.m. to 10.22 p.m.	Discussion on designing the prototype	Design the prototype and build the models			Started the implementation.	Satisfied.	S P Pitigala